A Bloomberg Professional Services Offering

YAS <GO>
Yield and Spread Analysis
Help Page

Enter YAS <GO>, then press <Help>

Date: 02/28/2024

Bloomberg

Contents

Definitions

For terms in blue throughout this document, see *Definitions on p.64.*

Frequently Asked Questions

Yield and Spread Analysis | YAS <GO>

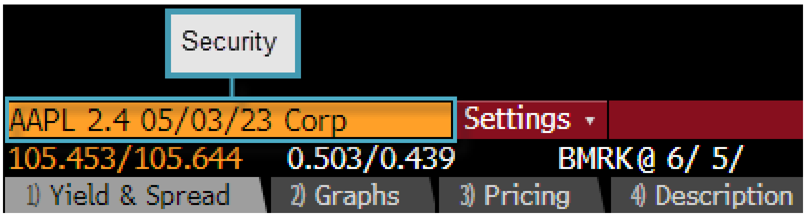
YAS <GO> integrates multiple fixed income functions, so you can evaluate the pricing, risk, and relative value of a fixed income security in one place. You can review descriptive information, analyze relative value, price a security based on custom inputs, and calculate yields using various market conventions, day types, and compounding methods.

Note: The features available in YAS may vary depending on the security you have loaded for analysis.

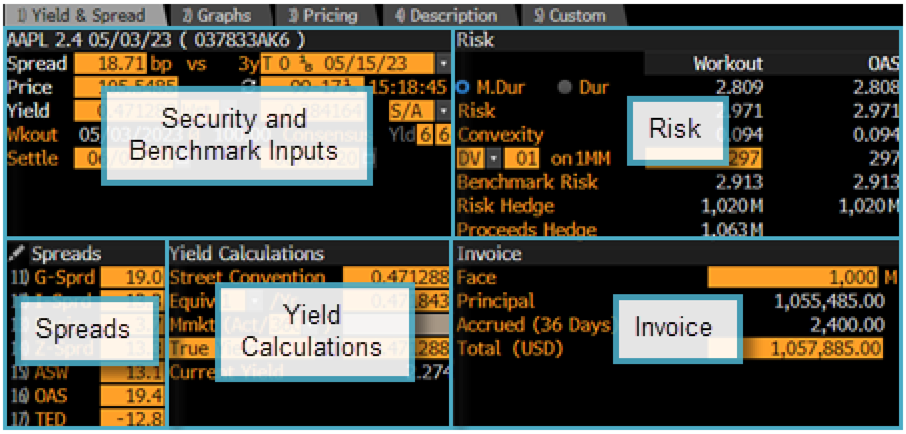
[Click through a tour of YAS](#)

Quick Start

1. Specify the **security** you want to analyze.



2. Customize the yield and spread calculations:



- **Security and Benchmark Inputs:** Choose a benchmark security, customize the spread, and input custom yield, price and settlement dates for the security under analysis and the benchmark.

- **Spreads:** Update the spreads, choose the curve from which the benchmark bond is selected for G-Spread and I-Spread calculations, and access related functions for spread analysis.

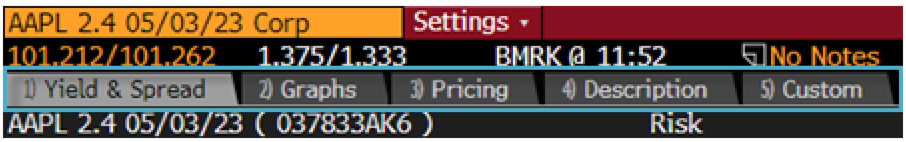
- **Yield Calculations:** Evaluate different yield convention calculations such as true yield, current yield, and annualized yield.

- **Risk:** Update the risk measure to hedge the security under analysis.

- **Invoice:** Customize the face amount to calculate the total amount required to settle a trade.

For more: Analyzing Yield/Spread.

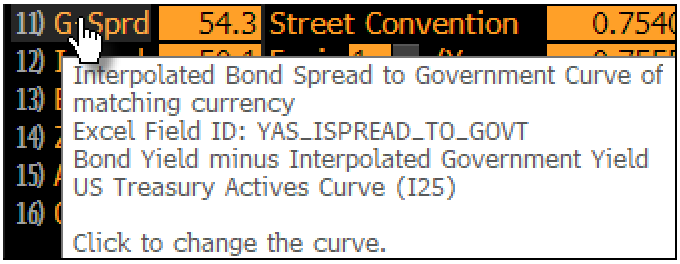
3. Select a tab to continue your analysis.



For more: Tab Descriptions.

Getting Hints as You Go

On any Bloomberg function, you can mouse over key items for more information:



Tab Descriptions

Depending on the type of security you have loaded, some of the following tabs may appear:

- **Yield & Spread:** Enter custom values to calculate spreads, yield, risk, cost of carry, and hedge ratios, so you can see if a security meets your investment criteria. You can choose the benchmark bond for spread calculations, and customize the benchmark's price, yield, and settle date. For more: Analyzing Yield/Spread.
- **Graphs:** Analyze historical spreads on four customizable charts, so you can quickly assess a security's performance. For more: Customizing Charts.
- **Pricing:** Monitor current market price data for the loaded security, including quotes from contributors and incoming messages, as well as TRACE data for US securities and exchange traded data for other securities, so you can determine the level at which the security is likely to trade. For more: Analyzing Pricing Data.
- **Description:** See descriptive information about the loaded security from the *Security Description* (DES) function for easy reference as you perform your analysis.
- **Custom:** Design a custom view with the information most important to your analysis. For more: Updating the Custom Tab.
- **Calls:** Calculate yields to predetermined call dates to analyze a callable security. For more: Yield to Call.
- **Puts:** Calculate yields to predetermined put dates to evaluate a puttable security. For more: Yield to Put.

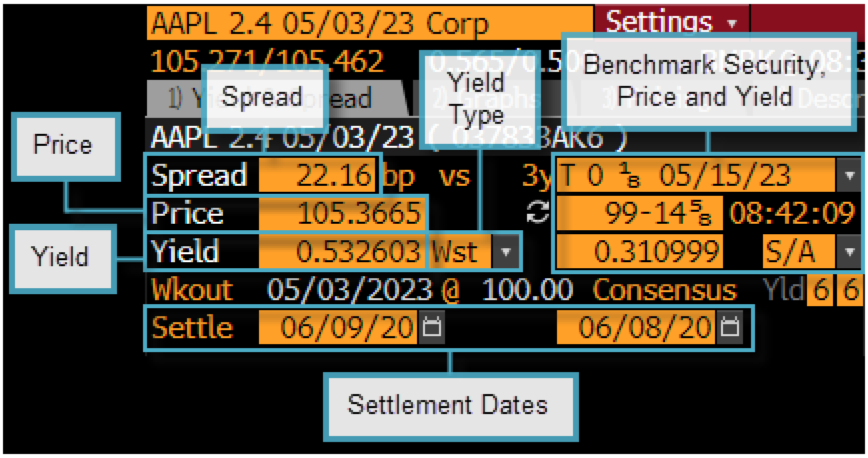
Workflows

Analyzing Yield/Spread

You can enter custom values to calculate spreads, yield, risk, cost of carry and hedge ratios to determine whether a security meets your investment criteria. The following topics explain how to perform a custom analysis in YAS.

Calculating Yield and Spread

You can calculate custom yields and spreads to analyze a security on every tab of YAS. You can choose a benchmark bond for spread calculations, and customize the benchmark's price, yield, and settle date.



- **Spread:** Enter a custom spread above benchmark to see a yield and price for the loaded security that reflects the specified spread.

  Note: You can click the refresh button next to the benchmark price to recalculate the spread based on the most recent benchmark price.
- **Price:** Specify a price for the loaded to security to see the yield you would earn at the specified price and the corresponding spread against the benchmark yield.
- **Yield:** Specify a custom internal rate of return for the loaded security to see the price at which you would earn the specified yield and the corresponding spread against the benchmark yield.
- **Yield Type:** Select a type (e.g., *Worst*) for the yield calculation. You can select *Custom* to specify a custom workout date and price. For more: Customizing Workout.
- **Benchmark Security, Price, and Yield:** Customize the benchmark bond used to price the loaded security and calculate spreads, and select a yield basis (e.g., *S/A* or semi-annual) for the benchmark and the loaded security. For more: Changing the Benchmark.
- **Settlement Dates:** Update the settlement date for the loaded security on the left or for the benchmark on the right to recalculate yields and spreads, as well as the values in the *Risk* and *Invoice* sections. You can calculate retroactive yields and spreads by entering a past settlement date in the *Settle* fields. For more: Measuring Risk/Invoice, Retroactive Yields/Spreads.

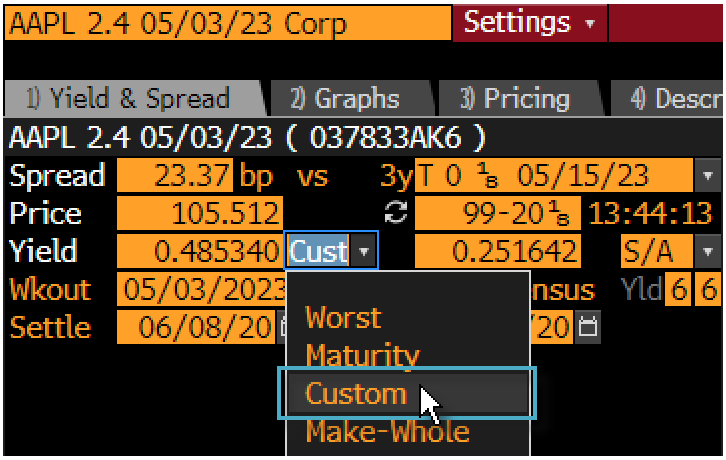
Customizing Workout

You can enter your own workout date and price to customize the yield calculation.

Steps:

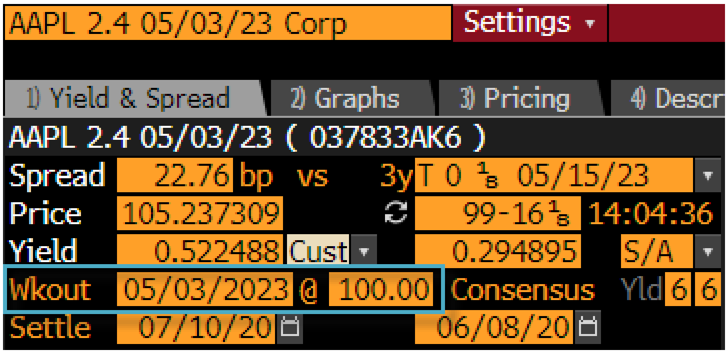
1. From the drop-down menu next to the *Yield* field, select **Custom**, then press <GO>.

   Note: The options that appear in the drop-down menu vary based on the type of security you are analyzing.

   The *Wkout* fields become active.

2. Update the *Wkout* date and/or price, then press <GO>.

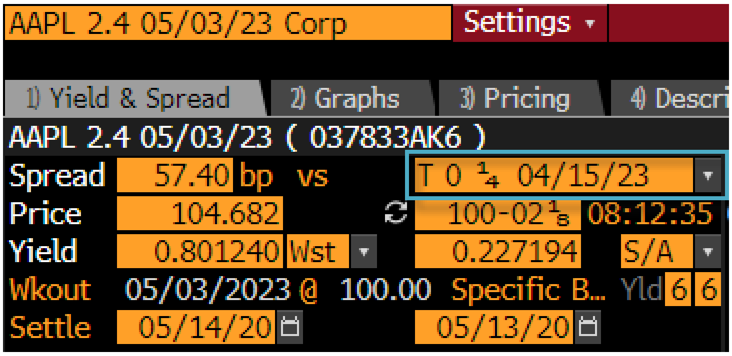
   The yield, price, and spread calculations update.

Changing the Benchmark

YAS launches with a default benchmark. You can change the benchmark to re-price the loaded security and recalculate spreads to a custom bond, swap, treasury curve, or to a local sovereign curve for emerging markets bonds.

Steps:

1. Select the benchmark you want to use for your pricing and spread calculations, then press <GO>:



- Enter the ticker symbol or security identifier for the new benchmark, then press <GO>.
- To apply a swap tenor as the benchmark, enter the swap rate ticker (e.g., *USSW10 <Crncy>*, *US0003M <Index>*).

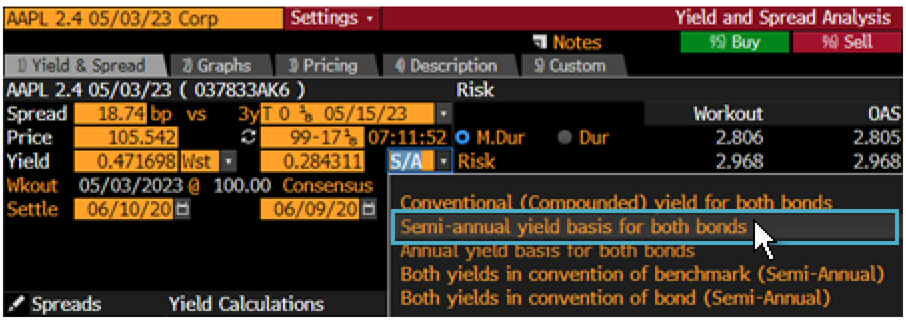
  Note: The current swap value is based on your settings in the *Swap Defaults* (SWDF) function. The spread is calculated by subtracting the swap value from the selected bond's workout yield. Only the spread to the benchmark swap updates. For more: SWDF Help Page.
- To apply a local sovereign curve in U.S. dollars to spread against an emerging markets bond, select a curve from the top-left section of the benchmark drop-down menu (e.g., *3mo B 0 09/3/20*).

  Note: To set the local USD sovereign curve as the default benchmark, select **Settings > YASD - YAS Settings** from the toolbar. Then, on the *Yield Curves and Other Tab*, select **Use USD sovereign curve for USD currency EM bonds**. You can also enter **YAS LOCAL** in the command line, then press <GO> to open YAS with a local sovereign bond in the benchmark field without changing YAS defaults.

2. If you want to save the benchmark bond, in the command line, enter **101**, then press <GO>.

   The benchmark bond is saved for the loaded security.

3. If you want to change the yield basis for the benchmark and the loaded security, make a selection from the yield basis drop-down menu (e.g., *Semi-Annual Yield Basis for Both Bonds*).

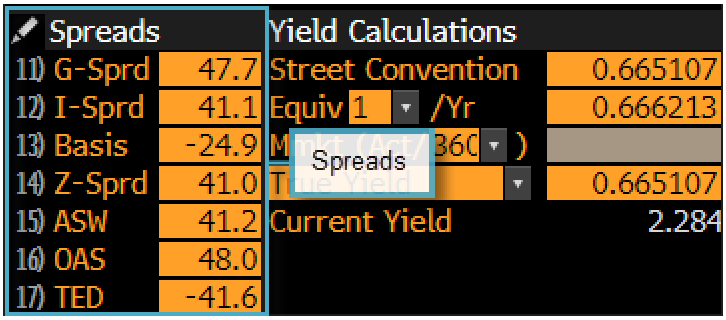


The yield basis updates.

Updating Spreads

The *Spreads* section displays the difference between the yield to the workout date and the yield interpolated to the matching point on various benchmark curves in the corresponding currency, so you can assess the relative value of the security under analysis.

The *Spreads* section appears on every tab of YAS.



You can update each spread field to analyze the spread values:

|Hint| You can change the list of spreads that appears by clicking the pencil icon.

Click	To
G-Sprd	Choose the curve from which the benchmark bond is selected for the G–Spread calculation.
I-Sprd	Choose the swap curve from which the benchmark is selected for the I–Spread calculation.
Basis	Update the CDS spreads used to calculate the basis.
Z-Sprd	Analyze the Z–Spread for the selected security from the *Asset Swap Calculator* (ASW) function. For more: ASW Help Page.
ASW	Analyze the ASW for the selected security in ASW.
OAS	Analyze the OAS for the selected security in the *Option-Adjusted Spread Analysis* (OAS1) function. For more: OAS1 Help Page.
TED	Analyze the Euro–Futures strip spread in the *Euro-Future Strip Hedge* (TED) function. For more: TED Help Page.

Note: YAS calculates G- and I-spread using the local sovereign benchmark and swap curves based on the currency of the security you are analyzing. You can override these default curves and currencies. For more: Overriding Curves/Currency.

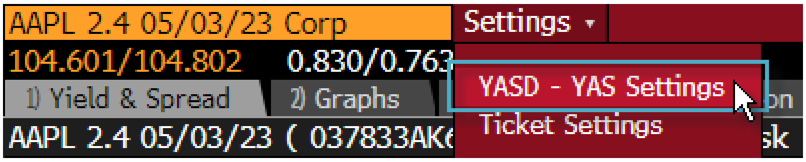
Overriding Curves/Currency

YAS calculates G- and I-spread using the local sovereign benchmark and swap curves based on the currency of the security you are analyzing. You can override the default calculation method by specifying your preferred curves on a currency-by currency basis to align your analysis with your trading currency.

Note: Swap curve overrides appear in your YAS settings automatically to apply risk-free rate (RFR) curves to I-spread calculations for USD, GBP, JPY, and CHF bonds by default. You can opt out of these default settings by creating your own swap curve overrides.

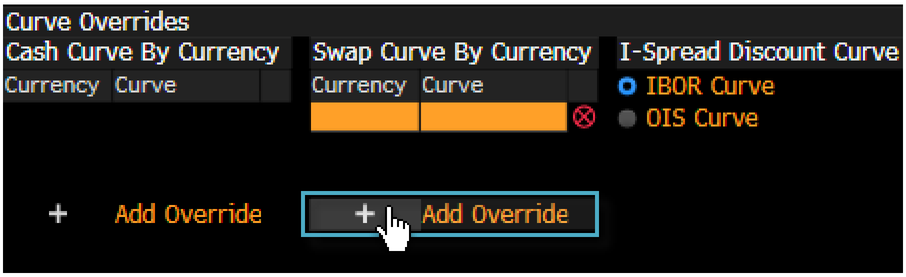
To create curve overrides:

1. From the toolbar, select **Settings > YASD - YAS Settings**.



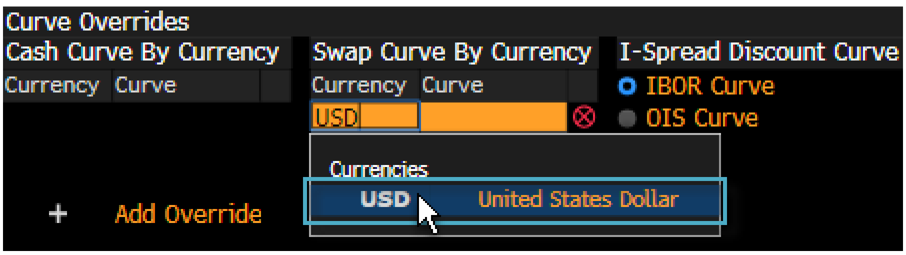
The *Yield and Spread Analysis Settings* (YASD) function appears. For more: YASD Help Page.

2. On the *Market Data* tab, in the *Curve Overrides* section, click the add icon corresponding to the type of curve you want to override (e.g., *Swap Curve By Currency*).

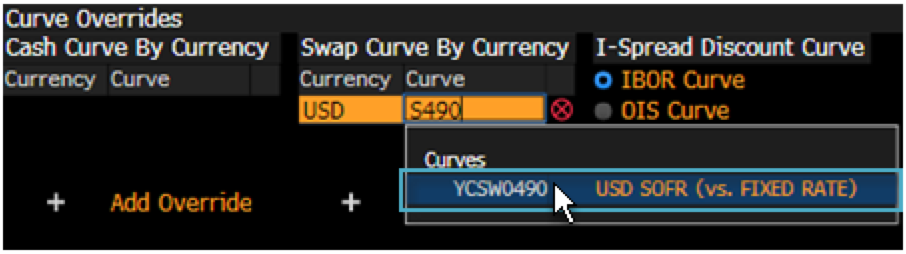


The *Currency* and *Curve* fields appear.

3. In the *Currency* field, start typing a currency (e.g., *USD*), then select it from the list that appears.



4. In the *Curve* field, start typing the curve you want to use (e.g., *USD SOFR (vs. FIXED RATE)*), then select it from the list that appears.



|Hint| You can also type a curve ID (e.g., *S490*) in the *Curve* field, then press <GO>. Curve IDs are available on the *Curve Finder* (CRVF) function. For more: CRVF Help Page.

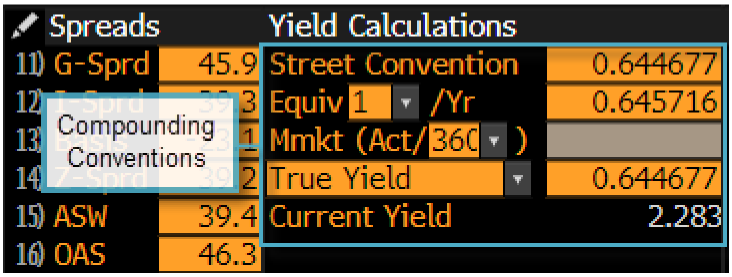
Your overrides are saved.

Note: You can remove an override by clicking the corresponding delete icon.

Compounding Conventions

The *Yield Calculations* section displays bond yields using multiple compounding conventions, so you can assess the relative value of the security under analysis. YAS displays the standard yield value for the security as the first yield value, followed by several equivalent yields.

The *Yield Calculations* section appears on every tab of YAS.



The following compounding conventions may appear:

Convention	Indicates
Street Convention	The yield calculated according to the conventions used in the market in which the bond was issued.
Equiv (number)/year	The street convention yield for the selected the compounding frequency. This method uses the same day type conventions and the actual dates of the security's cash flows.

Convention	Indicates
Mmkt (Act/)	The money market equivalent yield that equates a periodic coupon–paying security to a security that pays interest at maturity. Each cash flow is compounded from its coupon payment date to the next payment date and so on until the last payment date. The total is present valued back to the present date via simple interest.
True Yield	The yield calculated with coupon dates moved from a weekend or holiday to the next valid settlement date.
US Govt	For multi–coupon bonds, South African governments, and U.K. strips, the yield calculated using semi–annual coupons in all periods and an ACT/ACT day type.
Japanese Simple	The annualized cash flow as a percent of the original flat price. Using the simple interest basis, the effects of compound interest are ignored.
Current Yield	The annual rate of return based on the price. It is calculated as [(stated dividend or coupon * par amount) /price] * 100. This is the actual income rate of return, as opposed to the coupon rate expressed as a percentage.

Measuring Risk/Invoice

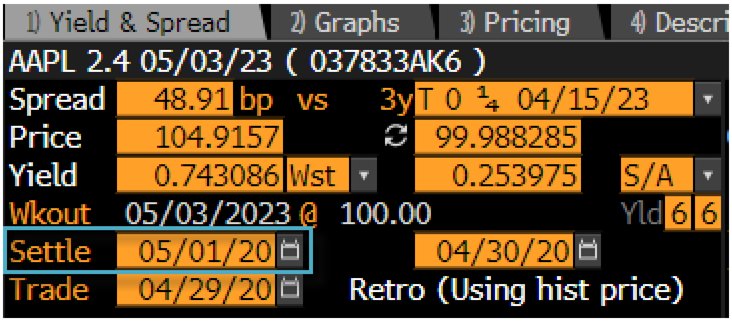
The *Risk* and *Invoice* sections of the *Yield & Spread* and *Pricing* tabs help you measure a security's risk and calculate the amount required to settle a trade, so you can determine whether a security matches your investment strategy.

- **Risk:** Hedge the security under analysis, using your selected benchmark as the hedge. If the corresponding security is option-free, the OAS risk measures are similar to the maturity measures. If the corresponding security is callable and in-the-money, traditional workout and OAS risk measures can produce different hedge measures.
- **Invoice:** Calculate the total amount required to settle a trade for a given face amount and settlement date.

Retroactive Yields/Spreads

You can calculate retroactive yields and spreads to analyze a security as of a prior date.

To calculate retroactive yields and spreads, in the *Settle* field, enter a past settlement date, then press <GO>.



The *Trade* date field appears with a *Retro (Using hist price)* message.

Note: YAS determines a trade date based on the retroactive settlement date. YAS imports an end-of-trade-day trade day price for the selected bond and the benchmark based on the PCS settings for each bond. YAS then calculates yields and all spreads based on the settlement dates and imported retroactive prices. When importing end-of-trade-day pricing, YAS searches for last, mid, ask, or bid prices, in that order. If no sourced price is available for the trade date, YAS defaults to a price of 99.99 for the bond, and a *Retro (Using default price)* message appears to the right of the *Trade* date field.

Analyzing Pricing Data

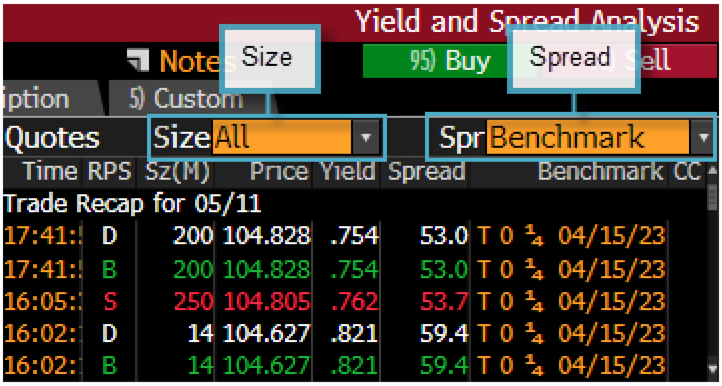
On the *Pricing* tab, you can track current market price data for the loaded security to help you the value a trade. The following topics explain how to customize and analyze the quote data that appears on the *Pricing* tab.

Customizing Quotes

In the *Quotes* section of the *Pricing* tab, you can customize the quotes that appear for a security, so you can focus on quotes that align with your investment criteria.

Note: The information that appears depends on the loaded security. For TRACE bonds, historical trade information for the security under analysis appears. For non-TRACE bonds, pricing information from the *All Quotes* (ALLQ) function appears. For more: TRAC <GO>, ALLQ Help Page.

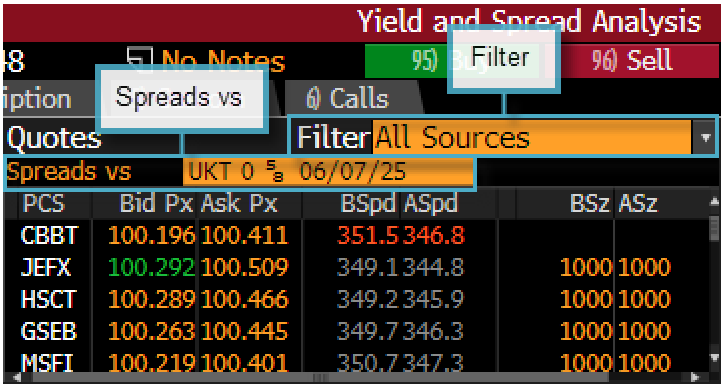
- To customize quotes for TRACE bonds:



- **Size:** Select the minimum size trade for which quote data appears.

- **Spread:** Choose the type of spread to display for each quote.

- To customize quotes for non-TRACE bonds:



- **Filter:** Choose whether to display quotes from all sources, executable quotes (*Executable Only*), or quotes from dealers with whom you are enabled to trade (*Enabled Only*).

- **Spreads vs:** Enter the benchmark bond to which the spreads are calculated, then press <GO>.

Updating QMGR Criteria

On the *Pricing* tab, you can update the search criteria driving the *Quotes Manager* section, so you can identify specific trade opportunities and pricing in the quotes you receive.

To update the *Quotes Manager* search criteria, click *Criteria Applied* pencil icon.

The *Quotes Manager: Edit Search* screen appears. For more: QMGR Help Page > Editing Search Criteria.

Related Analytics

In the *Quotes Manager* section of the *Pricing* tab, you can right-click a quote to access additional information, report a message mining issue, or communicate with the quote's dealer.

To access additional functionality for a quote, right-click a quote, then make a selection from the list that appears:

Note: The options that appear depend on the source of the quote.

- **Bloomberg Functions:** Analyze or trade the security in another Bloomberg Terminal® function (e.g., *All Quotes (ALLQ)*). For more on a function, see the function's Help Page.
- **Respond to Sender (MSG):** Communicate with the sender of the quote via the *Message* (MSG) function. For more: MSG Help Page.
- **IB Chat with Sender (IB):** Communicate with the sender of the quote via an Instant Bloomberg (IB) chat. For more: IB Help Page.
- **Report Mining Issue:** Report a parsing error for a quote to Bloomberg Message Mining. For more on message mining: BMIN <GO>.
- **View Original Message:** Display the original message containing the quote that was sent to your inbox or your MSG1 group.
- **View Offering Page:** Display the inventory page of the dealer who submitted the quote.

Quotes Manager Colors

In the *Quotes Manager* section of the *Pricing* tab, the entries in the *Src* column are color-coded to help you quickly see if the price is validated or if it comes from a dealer quote in the *All Quotes* (ALLQ) function.

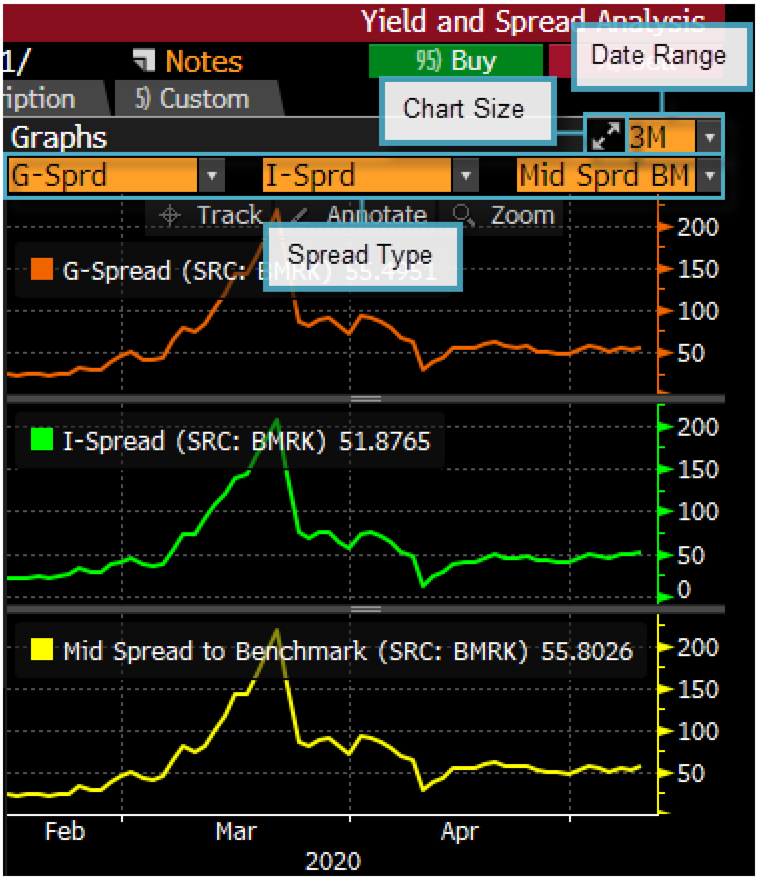
The following colors indicate the price status:

The color	Indicates
Green	The price is verified.
Blue	The price is not verified.
White	The quote is extracted from a pricing source in ALLQ.

For more: ALLQ Help Page.

Customizing Charts

On the *Graphs* tab, you can change the size, date range, and spread type plotted on each chart to customize your analysis.



- **Chart Size:** Expand the charts to the full screen width.
- **Date Range:** Select or enter the time period (e.g., *3M*) for each chart.
- **Spread Type:** Choose the type of spread (e.g., *G Spread, I Spread*) plotted on each chart.

Updating the Custom Tab

You can change the information that appears on the custom tab to focus on the data most important to your analysis. The following topics explain how to update the *Custom* tab.

Customizing Components

On the *Custom* tab, you can create a custom view, so you can focus on the data most important to your analysis.

Steps:

1. From any of the component drop-down menus, select the type of data you want to display (e.g., *DES (Description)*, *GIP I(Intraday Price Chart)*, *ALLQ (All Quotes)*, and *Rating (Bond Ratings)*).

The screen updates with data for the selected components. For more on each component: Definitions, Displaying YAS Activity.

2. If you want to save the custom view, in the command line, enter **99**, then press <GO>.

Your custom view is saved.

Displaying YAS Activity

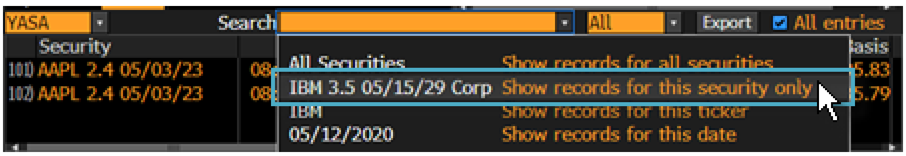
On the *Custom* tab, you can display a log of calculations you performed in YAS over the past 14 days, so you can keep track of bonds you have analyzed.

Steps:

1. From the component drop-down menu on the bottom-left of the *Custom* tab, select **YASA YAS Activity**.

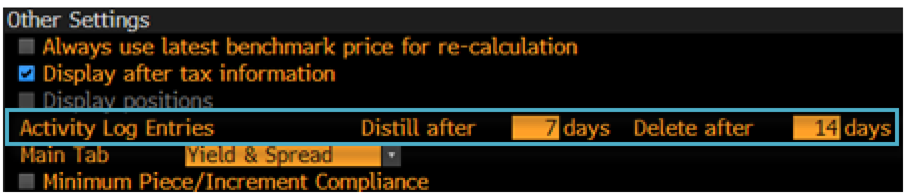
The *YASA* component appears across the bottom the screen with the 200 most recent entries.

2. If you want to filter the log for a specific security so you can display up to 2000 calculations for the preceding 90 days, from the search field, enter or select your filter criteria (e.g., *IBM 3.5 05/15/29 Corp*).



The activity log is filtered.

3. If you want to change the number of days information is retained in the activity log, from the toolbar, select **Settings > YASD YAS Settings**, then, on the *Yield Curves and Other* tab, update the *Activity Log Entries* fields and press <GO>:



- **Distill After:** Specify the number of days after which the activity log is distilled to retain only the last calculation of the day for each security.

- **Delete After:** Specify the number of days after which entries are deleted from the activity log. Activity is deleted automatically after 90 days unless a note is attached to the record.

Your activity log settings are updated.

Yield to Call/Put

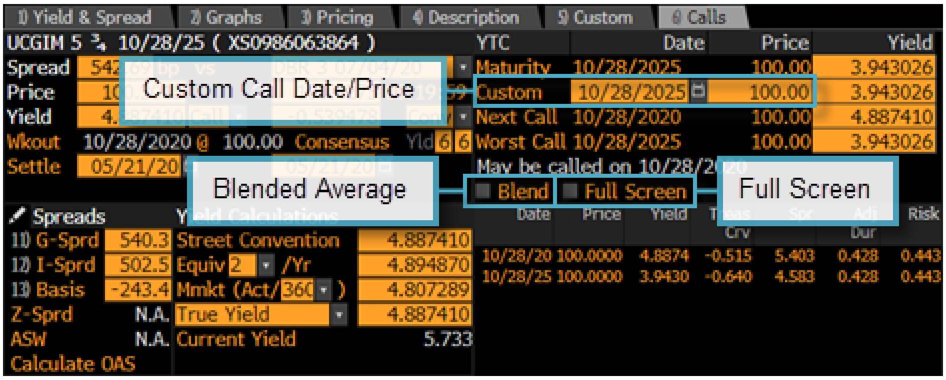
You can calculate yields to specific call or put dates to analyze callable or puttable securities. The following topics describe how to calculate yields on the *Calls* or *Puts* tab and how to analyze preferred securities in the *Yield to Call Analysis* (YTC) function.

Yield to Call

On the *Calls* tab, you can calculate yields to predetermined call dates, so you can measure a callable bond's value. You can see the different yield scenarios at each possible redemption date and calculate the blended average yield.

For U.S. dollar securities, yields are compared to the yields implied by the U.S. Treasury yield curve. For non-U.S. dollar securities, yields are compared to the yields implied by the corresponding government's yield curve as listed in the *Curve Finder* (CRVF) function. For more: CRVF Help Page.

Note: The *Calls* tab appears for callable bonds only.



- **Custom Call Date/Price:** Enter a custom call date and price.

  Note: The *Custom* date must be on or before the Worst Call date.

- **Blended Average:** Analyze the blended average yield. Once you select *Blend*, additional *% Called* fields appear and a *Risk-Weighted Blended Average Yield to Custom* field appears with the blended average yield.

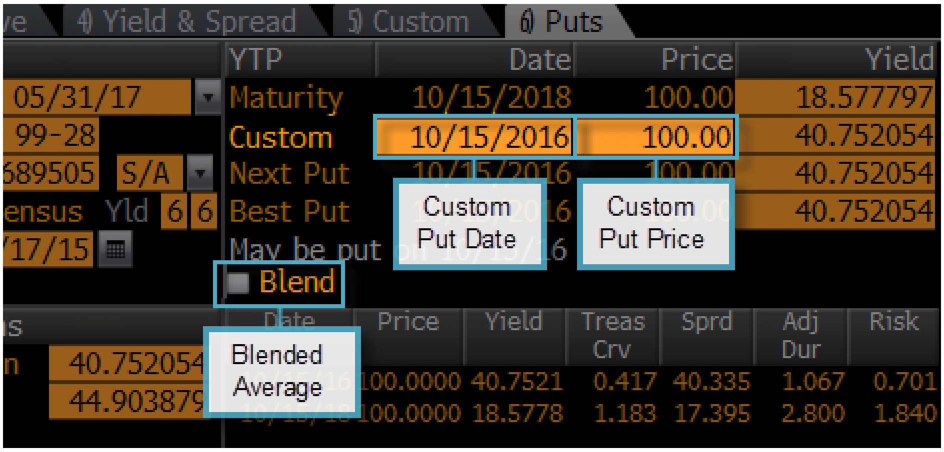
- **Full Screen:** Hide the fields on the left side of the screen and extend the yield-to-call fields across the *Calls* tab.

Yield to Put

On the *Puts* tab, you can calculate yields to predetermined put dates, so you can measure a putable bond's value. You can see the different yield scenarios at each possible redemption date. You can also analyze the blended average yield.

For U.S. dollar securities, yields are compared to the yields implied by the U.S. Treasury yield curve. For non-U.S. dollar securities, yields are compared to the yields implied by the corresponding governments yield curve as listed in the *Curve Finder* (CRVF) function. For more: CRVF Help Page.

Note: The *Puts* tab appears for putable bonds only.



- **Custom Put Date/Price:** Specify a custom put date and price to calculate yield.

  Note: The *Custom* date must be on or before the *Worst Call* date.

- **Blended Average:** Analyze the blended average yield, along with the *Risk-Weighted Blended Average Yield to Custom* and *% Called*.

Analyzing YTC for Pfds

For preferred securities, you can calculate yields to predetermined call dates in the *Yield to Call Analysis* (YTC) function, so you can value the security at each date based on your own redemption assumptions.

For U.S. dollar securities, yields are compared to the Treasury yield curve. For non-U.S. dollar securities, yields are compared to the yields implied by the corresponding government yield curve.

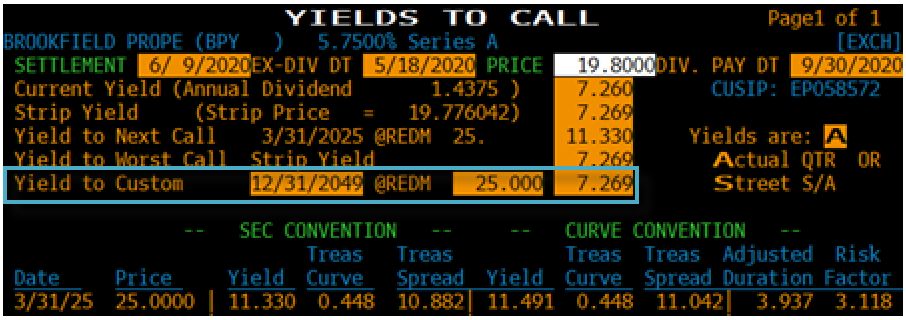
Note: You can leverage the range of tools available in YAS to analyze preferred securities, including yield to call analysis. For more: Yield to Call.

1. In the command line, enter YTC for a preferred security (e.g., *BPY 5.75 PERP <Pfd> YTC*), then press <GO>.

   YTC appears with the security loaded.

2. At the top of the screen, update the highlighted fields. For example, if you want to calculate the yield on a specific date, update the Yield to Custom and adjacent @REDM fields.

   For more on the fields that appear: Definitions.



The yields at the top of the screen and in table at the bottom of the screen update based on your inputs.

Note: If the security and the comparable government curve have different compounding frequencies, two sets of values appear for each yield, curve, and spread in the yield table section. *Sec Convention* displays the yield and curve values using the compounding frequency of the security. *Curve Convention* displays the yield and curve values using the compounding frequency of the yield curve.

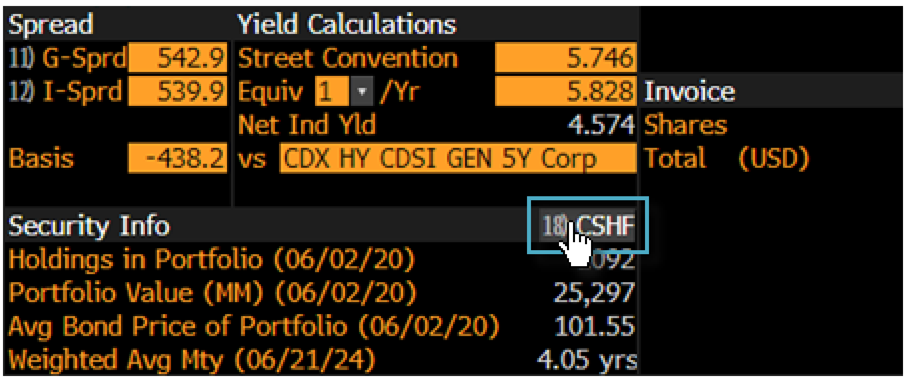
Accessing ETF Cash Flows

To facilitate the management and trading of fixed income exchange traded funds, an advisory group consisting of Bloomberg, BlackRock, and State Street Global Advisors developed a market convention, known as the Aggregate Cash Flow method (ACF), for calculating yield, spread, and duration for fixed income ETFs.

On the *Yield & Spread*, *Pricing*, and *Yields* tabs, you can access the *Cash Flow Analysis* (CSHF) function to see aggregated period-by-period cash flows of fixed income ETFs for insight into the ACF method.

Note: Currently, YAS supports calculations using the ACF method for single-currency corporate and government fixed income ETFs (e.g., TLT US <Equity>). For more: Fixed Income ETFs.

To see aggregated period-by-period ETF cash flows, in the *Security Info* section, click the **CSHF** button.



CSHF appears with information for the fixed income ETF. For more: CSHF Help Page.

Ticketing a Trade

You can move seamlessly from yield/spread analysis to submitting a trade ticket for the loaded security for a streamlined workflow.

To submit a trade ticket, click the **Buy** or **Sell** button.

The trade ticket corresponding to your ticket settings appears. For more: Ticket Settings.

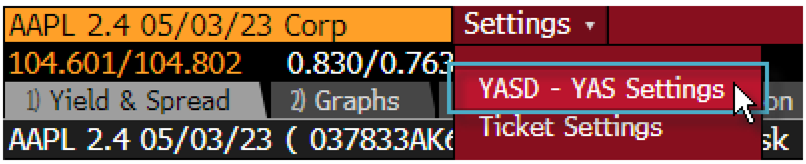
Settings

You can update settings to customize the features and ticketing options that appear by default when you run YAS. The following topics explain how to update YAS settings.

YAS Settings

You can access the *YAS Defaults* (YASD) function to customize the price type (bid/ask/mid), yield precision (number of decimal places), price rounding, benchmark selection, yield curve, and display settings that appear in YAS by default.

To update default settings, from the toolbar, select **Settings > YASD - YAS Settings**.

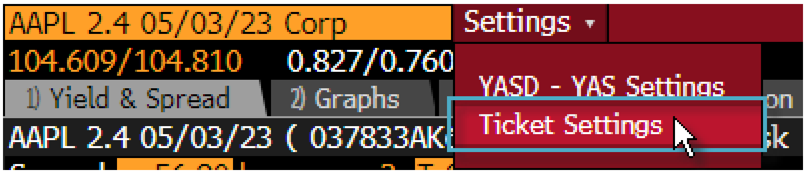


YASD appears. For more: YASD Help Page.

Ticket Settings

You can designate a specific type of trade ticket to appear automatically when you click the *Buy* or *Sell* button based on your trading workflow.

1. From the toolbar, select **Setting** > **Ticket Settings**.



The *Ticket Settings* window appears.

2. Select a trade ticket type:

- **Trading Systems:** Submit trade tickets in the *Trading Solutions Buy Ticket* (B) or *Trading Solutions Sell Ticket* (S) functions. This option is only available for Trading Systems users. For more: B Help Page, S Help Page.
- **Electronic Trading:** Enter trade tickets in the *Buy Inquiry Ticket* (BYI) and *Sell Inquiry Ticket* (SLI) functions. For more: BYI Help Page, SLI Help Page.
- **Voice (BXT/SXT):** Submit trade tickets in the *Buy Ticket* (BXT) and *Sell Ticket* (SXT) functions. For more: BXT Help Page, SXT Help Page.
- **STW Sales Inquiry:** Launch a price or spread sales inquiry ticket for the loaded bond, pre-filled with your bond amount, settlement date, price, and benchmark bond details from YAS.

  Note: You must be enabled for the Sales Inquiry Workflow to access STW Sales Inquiry tickets from YAS. For more: RFQX Help Page.
- **Always Choose My Ticket Type:** Select your ticket type when you click the *Buy* or *Sell* buttons.

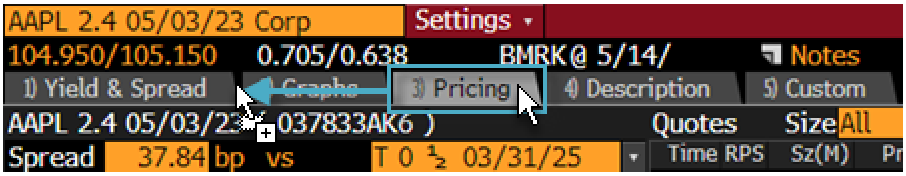
3. Click the **Close** button.

Your default ticket type is saved.

Re-Ordering Tabs

You can change the order in which YAS tabs appear to customize your default view.

To re-order tabs, select a tab, then drag it to a new position.



Your new tab order is saved. The tab appears in its new position the next time you run YAS.

Notes About a Security

You can create and review notes about the loaded security to record your ideas or to share insights with your colleagues. The following topics describe how to create and access notes about a security from YAS.

Accessing Notes

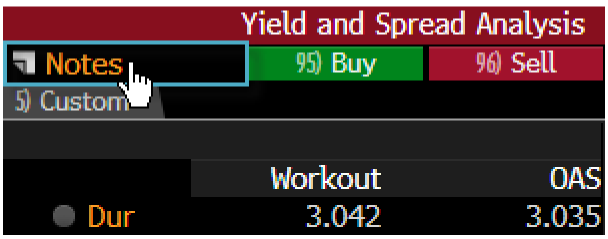
You can review and enter notes about the loaded security for yourself or your colleagues, so you can instantly and transparently share trading ideas and insights. The notes are saved and stored in the *Notes* (NOTE) function.

The type of *Notes* button that appears reflects the availability of notes on the loaded security. For more: Notes Button.

|Hint| YAS aggregates notes based on the settings you select in NOTE. For more: NOTE Help Page > Search Notes by Issuer or Entity.

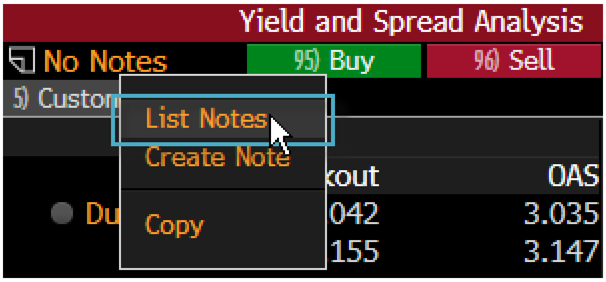
To add or review notes for the loaded security:

1. Click the **Notes** button at the top of the screen.



   NOTE appears. Depending on your YAS settings, NOTE may appear in-panel or in a *Collaboration Center* Launchpad window. If you do not have any notes for the loaded security, the *Create* screen in NOTE appears. For more: Note Help Page > Working With Notes.

2. If you do not have any notes on the loaded security, or if you want to browse your notes for related securities, right-click the **Notes** button, then select **List Notes**.



The search screen in NOTE appears, where you can adjust the filters to find notes on a different security.

Notes Button

The type of *Notes* button that appears on YAS reflects the availability of notes on the loaded security.

For more: Accessing Notes.

Notes Button	Indicates
No Notes	You do not have any personal or shared notes on the loaded security.
Notes	You have personal or shared notes on the loaded security.
New Notes	You have new (unread) shared notes on the loaded security that were updated or shared with you within 24 hours.

Note: YAS aggregates notes based on the settings you select in the *Notes* (NOTE) function. For more: NOTE Help Page > Search Notes by Issuer or Entity.

Reference

Calculations

Calculations by Security Type

This section includes the calculations used in yield analysis for specific security types. The calculations are listed in alphabetical order.

Bonds in Default

This section describes the calculations used in yield analysis for bonds in default.

A bond is said to be in default when it fails to fulfill its contractual obligations, such as the delay or failure to make timely payments of interest or principal.

Projected First Payment

This section assumes that the security resumes its scheduled interest by a specified date.

Note: No provision is made for any back interest payment, as it is assumed that the quoted price includes all interest.

- **If 1st Pay Is:** The first payment date.
- **Yield:** The yield, calculated to the IF 1ST PAY IS date.

Brady Bonds

This section describes the calculations used in yield analysis for Brady bonds.

For more on Brady bonds: Definitions.

Types of Brady Bonds

- **Par and Discount Bonds:** These bonds can be either fixed- or floating rate collateralized bonds (the most common type). Par bonds are issued for the original face value of the loans for which they are exchanged. However, par bonds have a fixed interest rate that is below the market rate (at the time of issuance). Discount bonds have a floating market interest rate and are issued at a discount to the original face value of the original loan. In general, both types have their principal at maturity collateralized by U.S. Treasury zero-coupon bonds. Both types of bonds usually possess additional collateral in the form of a cash amount that can be used to pay interest for a specified number of months (usually 12 to 18) in the event that the sovereign country misses an interest payment. Therefore, this collateral provides a "rolling" interest payment protection that can be applied to guarantee the nearest dated interest payments of the instrument.
- **Debt Conversion Bonds (DCBs), Front Loaded Interest Reduction Bonds (FLIRBs), and New Money Bonds (NMBs):** These bonds are amortizing bearer instruments, characterized by shorter maturity and average life than the par and discount bond types. They also lack collateral securing the principal. Only FLIRBs carry collateral securing interest payments (usually for 12 months). FLIRBs are bonds that carry a fixed, below-market interest rate that rises incrementally over the initial five to seven years of the life of the bond, which is then replaced by a floating rate coupon for the remaining life of the bond.

Analyzing Brady Bonds

Brady bonds can be analyzed by independently valuing each component. Brady bonds consist of some or all of the following:

- A redemption value, representing the principal of the bond, that can be collateralized, paid out at maturity, or sunk during the bond's life.
- If the bond has an interest guarantee, it can include a series of collateralized rolling interest payments.
- A series of unsecured interest payments, which are the responsibility of the sovereign nation.

Modified Cashflow Model

There are different analytical approaches to valuing the yield on Brady bonds. This has led to frustration in the market place, as these methods give different results. Bloomberg, working in conjunction with analysts at JP Morgan Securities Inc., has adopted the JP Morgan Modified Cash Flow (MCF) approach as the standard for analyzing Brady bonds.

Note: You can invoke this methodology in the *Bond Calculation Override* (COVR) function. Enter "Y" in the *Use Modified Cashflow Model* field. For more: COVR Help Page.

The MCF Model uses three yield curves:

- The current U.S. Treasury issues, assumed to consist of par bonds
- The U.S. dollar swap curve with points to match the active U.S. Treasury curve, used as a proxy for the LIBOR swap curve
- U.S. Treasury strip rates

The unique characteristic of the MCF model is its treatment of the rolling interest guarantee (RIG) feature contained in many Brady bonds. RIG is divided into two portions:

- The benefit given to those initial coupons currently covered by the funds contained in the interest collateral account
- The benefit that is given to later coupons in the cash flow stream that are not currently covered

The initial coupons are free of default risk, since they are covered by the guarantee no matter what happens. Coupons can be valued at swap zero rates. The analysis calculates the present value of the interest guarantee by discounting the sovereign cash flow immediately guaranteed at the appropriate swap implied zero rates, derived from the active U.S. dollar swap curve. The remaining interest cash flows, although not currently covered by collateral, still benefit from the RIG. The RIG effectively shortens the length of time over which these payments are subject to sovereign risk. The rolling nature of the RIG analysis is handled by discounting them at the appropriate implied forward swap zero rates over the rolling guarantee period.

Brady bonds may also be subject to a guarantee on the principal payment. If such a guarantee exists, the model values it by discounting the principal at the derived Treasury strip rate corresponding to the maturity date of the security.

The sovereign risk cash flow stream resulting from removing the RIG value and the principal guarantee can be viewed as the original bond's stream discounted at U.S. Treasury rates that are shifted forward. Several measurements have been developed for analyzing these securities.

Initial Coupon Guarantee Model

One of the first models developed for analyzing Brady bonds is the Initial Coupon Guarantee model. Most investors consider this to be an inferior model to the Modified Cashflow Model. In this model, the guarantee is applied to only the nearest coupons to the settlement date.

You can invoke this methodology in the *Bond Calculation Override* (COVR) function. Enter "N" in the *Use Modified Cashflow Model* field to indicate that you want YA to use the Initial Coupon Guarantee Model. For more: COVR Help Page.

In this model, a Brady bond consists of some or all of the following:

- A redemption value, representing the principal of the bond, may be fully or partially collateralized, paid out at maturity, or sunk during the bond's life.
- A redemption value, representing the uncollateralized portion of the principal of the bond, paid out at maturity or sunk during the life of the bond. This is the responsibility of the sovereign nation.
- A series of collateralized rolling interest payments, of a specified time span, applied to the nearest future interest payment as time progresses (i.e., after the current period containing settlement). For example, for a bond paying on 2/15 and 8/15

and a settlement of 6/1/98, the holder of the security receives the interest payment on 8/15/98 and the collateralized payments are removed beginning with the 2/15/99 payment.

- A series of unsecured interest payments, which are the responsibility of the sovereign nation.

To value a Brady bond, you must value each piece. To get a clear picture of market perception for a country's risks, you must look at the pure sovereign risk, which is composed of the debt service cash flows subject to payment by the sovereign nation. This is done as follows:

- The collateralized principal is valued by discounting it by the appropriate U.S. Treasury rate (or other government yield rate, if so collateralized). This rate can be an interpolated yield curve value or a derived strip curve value.

- The collateralized rolling interest payments are discounted by the appropriate U.S. Treasury rate (or other government yield rate, if so collateralized). This rate can be an interpolated yield curve value or a derived strip curve value.

- Subtract the present values of the collateralized principal and rolling interest guarantees from the current price of the instrument. This is the strip price.

- Value the remaining cashflow (the unsecured cashflow) and find the strip yield that corresponds to the calculated strip price.

Brazilian Government Debt

This section describes the calculations used in yield analysis for Brazilian government debt.

National Treasury Notes Series D (NTN-D) and Central Bank Notes Special Series (NBC-E): YIELD CALCULATIONS (bonds issued prior to Sept. 1, 2000 - Calc Type 714)

- **Unitary Price (US$ Indexed):** The price updated by the dollar variation. Brazilian traders use this price to sell or buy a note after agreeing on the yield. It is calculated as:

  PU$ = Face * Q%

  where:

  PU$ = Unitary price updated by the PTAX variation

  Face = Face value updated by the PTAX variation

  Q% = Percentage of the market quote

- **Market Yield to Maturity:** The yield of the note, according to the market that trades notes based on their yield. Defaults to prices provided by contributors. This is the internal rate of return (IRR) of the note, expressed on an exponential annual basis.

  To change your price contributor, enter **PCS** <GO>.

- **Unitary Price:** The price calculated using a discounted cash flow based on the coupon and the yield (non-dollar indexed), calculated as:

  PUc = Par * Q%

where:

PUc = Unitary price (including accrued interest)

Par = Par value

Q% = Market quote

- **Market Quote:** The price calculated using a discounted cash flow based on the coupon and the yield, expressed as a percentage of the face value. The coupon is expressed on an exponential annual basis, using an ACT/360 day count, and paid semiannually. The yield (internal rate of return) is also calculated on an exponential annual basis using an ACT/365 day count. You can measure your cash flows based on this value in the *Cashflow Analyzer* (PF) function. For more: PF Help Page.

- **%PTax (Dollar Rate):** The denomination of the dollar price calculated by the Brazilian Central Bank. The variation is calculated from the day prior to the issue date to the date before the settlement/maturity of the note. You must use the ask price from the day prior to the issue date. The BZFXPTAX <Index> is used to calculate this rate:

  PTAX = PTAXt-1/PTAXt0

  where:

  PTAX = Variation of the US$ since the issue date

  PTAXt-1 = Ask price of PTAX on the day prior to the settlement/maturity date

  PTAXt0 = Ask price of PTAX on the day prior to the issue date

- **Treasury Quote:** Appears for NBC-Es and NBC-Fs. The government quote updated to include the rate given by the Central Bank at the issue. This price does not include embedded dollar variations.

- **Treasury Quote @ Issue:** The government quote from the auction date (refers to how much the note cost and the price at issue).

- **Treasury Yield to Maturity:** The government yield at auction. If you buy a government note at auction, this is the yield the government pays you.

- **Daily Coupon:** The daily coupon calculated for each period, using an ACT/365 day count. Each period begins after the last coupon payment. If the coupon/maturity date falls on a weekend or holiday, bump the number of days to the next business day. The daily coupon is calculated as:

  DC = ((C/100)+1)^(ACT/365), where:

  C = Coupon

- **PTax @ Issue:** The Brazilian dollar average rate, calculated by the Brazilian Central Bank at the end of the day, one day prior to the issue date. For more information, enter **BZFXPTAX <Index> DES** <GO>.

- **PTax @ Settle:** The Brazilian dollar average rate, calculated by the Brazilian Central Bank at the end of the day, one day prior to the settlement/maturity date. For more information, enter **BZFXPTAX <Index> DES** <GO>.

- **Md:** The multiplier factor for the first coupon, according to the methodology established by Circular 2.960 on Jan. 19, 2000. This multiplier represents a pro-rata adjustment in the first coupon when the note is not issued on a date equal to the maturity date.

National Treasury Notes Series D (NTN-D) and Central Bank Notes Special Series (NBC-E): YIELD CALCULATIONS (bonds issued after Sept. 1, 2000 - Calc Type 1036)

- **Nominal Market Yield to Maturity:** The internal rate of return (IRR) of the note expressed in a linear annual basis. Defaults to prices provided by contributors.

  To change your pricing contributor, enter **PCS** <GO>.

- **Unitary Price (US$ Indexed):** The price updated by the dollar variation. Brazilian traders use this price to sell or buy a note, after they have agreed upon the yield. Unitary price is calculated as:

  PU$ = Face * Q%

  where:

  PU$ = Unitary price updated by the PTAX variation

  Face = Face value updated by the PTAX variation

  Q% = Percentage of the market quote

- **Exponential Mkt Yield to Maturity:** To discount the cash flows, convert the linear yield into an exponential yield using a 30/360 day count.

- **Market Quote:** The price is calculated using a discounted cash flow based on the coupon and the yield expressed as a percentage of face value. The coupon is expressed on a linear annual basis using a 30/360 day count and paying semi-annually.

  To measure cash flows based on this value, enter **PF** <GO>.

- **%PTAX Variation:** The denomination of the dollar price calculated by the Brazilian Central Bank. The variation is calculated from the day prior to the base date until the day prior to the settlement/maturity of the note. To calculate %PTAX variation, use the ask price from BZFXPTAX <INDEX>:

  PTAX = PTAXt-1/PTAXt0

  where:

  PTAX = variation of the US$ since the issue date

  PTAXt-1 = Ask price of PTAX on the day prior to the settlement/maturity date.

  PTAXt0 = Ask price of PTAX on the day prior to the issue date.

- **PTax @ Base date:** Appears for NBC-Es and NTN-Ds. The Brazilian dollar average rate calculated by the Brazilian Central Bank at the end of the day, one day prior to the base date. PTax @ Base date is established at the auction announcement. For more information, enter **BZFXPTAX <Index> DES** <GO>.

- **PTax @ Settlement:** Appears for NBCEDs and NTN-Ds. The Brazilian dollar average rate calculated by the Brazilian Central Bank one day prior to the settlement/maturity date. For more information, enter BZFXPTAX <INDEX> DES <GO>.

- **Daily Coupon:** The daily coupon calculated for each period using a 30/360 day count. Each period begins after the last coupon payment. The daily coupon is calculated as:

  DC = ((C/100)+1)^(ACT/365)

- **Coupon Value by Ptax%:** The DAILY COUPON updated by the %PTAX VARIATION.
- **To:** The maturity/workout date.

- **Face (US$ Indexed) R$:** The total face value updated by the dollar variation, expressed in Brazilian real. The face is calculated as:

  Face = 1,000 * PTAX

  where:

  Face = face value updated by PTAX variation

  1,000 = face value of each note in Brazilian reals (BRL)

  PTAX = variation of the US$ since the issue date

- **Net Principal (US$ Indexed):** The gross principal less the accrued interest, using dollar variation.

- **{XX} Days Accrued:** The interest accrued after {XX} days over the FACE (US$ INDEXED) beginning from the last coupon payment. The interest accrued is calculated as:

  IA = Face * DC

  where:

  IA = interest accrued

  Face = Face value updated by PTAX variation

  DC = daily coupon

- **Gross Principal:** The par value updated by dollar variation multiplied by the amount of notes. This amount to the total value, in Brazilian reals, to be paid at the trade. Gross principal is calculated as:

  Gross = Face * Q * amount of notes

  where:

  Face = face value updated by PTAX variation

  Q = percentage of the market quote

- **Coupon Value:** The daily coupon multiplied by the par value.

- **Coupon (US$ Indexed):** The daily coupon multiplied by the par value plus the dollar variation.

- **Market Return:** Appears for NBCEDs and NTN-Ds. The return the market expects on the issue, based on the market yield.

- **Government Return:** Appears for NBCEDs and NTN-Ds. The return the market expects on the issue, based on the government yield.

Central Bank Bills (BBC), Treasury Bills (LTN) (Calc Type 712)

BBCs and LTNs are yield quoted and traded with an Over rate per month. The Over rate is a typical Brazilian calculation, which uses the number of business days of the month. BBCs and LTNs are traded at a discount over the par value, in reals.

- **Unitary Price:** The price of the bond. Refers to the par value traded at a discount. The unitary price is calculated as:

  PU = PV/((i/100)+1)^(BDM/252)

  where:

  PV = Par Value

  i = yield

  BDM = business days between settlement and maturity. For further information, enter CDR BZ <GO>.

- **Yield to Maturity Annualized {XX} Business Days:** The yield annualized by a period of {XX} business days (using the over rate methodology), calculated as:

  Oy = {[(O/3000 + 1) ^ 252] - 1 } x 100

  where:

  Oy = Over rate per year

  O = Over rate per month

- **Yield to Maturity (Non-Annualized):** The yield of the bond from the issue date to the maturity date, expressed as an over rate per month (using the business days).

- **Over Rate:** The price of the bond, expressed as an over-night rate, according to the security's traded value in the market. The over rate is calculated as follows:

  O = {[(1000/PU) ^ (1/BDM)] -1 } x 3000

  where:

  PU = Unitary price

  O = Over rate per month

  BDM = Business days to maturity

- **Yield Annualized x SELIC:** The variation, in percentage, between the bond and the SELIC rate. The SELIC rate is an average interbank lending rate for operations guaranteed by government bonds.

- **Principal:** The "discounted" par value, based on the yield. BBCs and LTNs have a par value of R$1,000, which is paid at the redemption date.

Brazilian Floating Rate National Treasury Bills (LFT)

LFTs are Brazilian floating-rate bills issued by the Brazilian National Treasury. LFTs yield the SELIC rate plus/less a premium or discount.

- **Market Quote:** The price calculated using a discounted cash flow based on the coupon and the yield, expressed as a percent.

  To measure your cash flows based on this value, enter **PF** <GO>.

- **Price Per Note:** The price calculated using a discounted cash flow based on the coupon and the yield (non-dollar indexed):

  PN = Adj Par * Q%

  where:

  PN = Price per note (including accrued interest)

  Adj Par = Par value adjusted by Selic

  Q% = Market quote

- **SELIC Accumulated (Factor):** The accumulated factor of the SELIC rate since the issue of the bond.
- **Par Value Adjusted by Selic:** The par value of the bond (R$1000.00 per note) adjusted by the accumulated SELIC, calculated as PAR VALUE * SELIC ACCUMULATED (FACTOR).
- **Annualized Selic:** The annualized SELIC rate is calculated using 252 business days per year.
- **Amount of Notes:** The number of bonds.
- **Par Value + Selic:** The par value adjusted by the accumulated factor of the SELIC rate.
- **Total R$:** The total value of the invoice in Brazilian real.
- **Quote @ Issue:** The market quote for the bond at the time of issue.
- **Premium/Discount @ Issue:** The positive or negative difference between the bond's market price at issue and its par value.

National Treasury Fixed Rate Bond Series F (NTN-F):NOTA DO TESOURO NACIONAL SERIES F (Calc Type 1309)

Brazilian fixed coupon rate bonds. This bond is traded dirty and at unitary price. The day count used is business days/252 business days. The par value of each unit is worth BRL 1,000.00. Dirty price means that accrued interest is included in the unitary price. The total payment invoice is the unitary price multiplied by the number of bonds.

- **Unitary Price:** The current price of the security, calculated using a discounted cashflow based on the coupon and the yield. The bond is traded dirty and in unitary price. Each unit is worth BRL 1,000.00. Dirty price means that accrued interest is included in this unitary price.
- **Yield to Maturity Annualized XX Business Days:** The yield annualized by a period of {XX} business days (using the over rate methodology), calculated as:

  Oy = {[(O/3000 + 1) ^ 252] - 1 } x 100

where:

Oy = Over rate per year

O = Over rate per month

- **Coupon:** The bond coupon.

  Note: Coupons are calculated on an exponential basis.

- **Clean Price (%):** The security's price minus the accrued interest, expressed as a percentage.

- **Dirty Price (%):** The full price of the security including accrued interest.

National Treasury Inflation-Linked Bond Series (B) (NTN-B): NOTA DO TESOURO NACIONAL SERIES B (Calc Type 1193)

Brazil Inflation linked securities, linked to the BZPIIPCA Index. Calculations are based on the number of business days in a year being equal to 252 days. These bonds are dirty and issued in unitary price, i.e., quoted in units. The par value of each unit is worth BRL 1,000.00. Dirty price means that accrued interest is included in the unitary price. The Total payment invoice is the unitary price multiplied by the number of bonds. The Par Amount is adjusted by index variation.

- **From Issue to Maturity:** The number of actual days and the number of business days from the issue date to the maturity date.

- **Quote%:** The quote percentage of par value (updated).

- **Price:** The current price of the security, calculated using a discounted cash flow based on the coupon and the yield. The security is linked to the index variation and is traded dirty and in units. Each unit is worth BRL 1,000.00. Dirty price means that accrued interest is included in the unitary price.

- **IPC-A @ Base Date:** The index value in the base date.

- **IPC-A:** The index value in the current date.

- **IPC-A Assumption:** The index assumption, published by Associacao Nacional das instituicoes do Mercado Aberto (ANDIMA).

- **Index Variation:** The index variation, which is based on the IPC-A assumption, calculated exponentially and taking into consideration the current and passed business days, IPC-A base date, and value.

- **Curr. Month Bus Days:** The date range of the current month and the corresponding number of business days passed in the current month.

- **Bus. Days Passed:** The date range of the current month business days passed and the corresponding number of business days passed.

- **Par Value:** The par value of the bond (R$1000.00 per note).

- **Principal:** The discounted par value, based on the yield, which amounts to the clean price multiplied by the amount of notes.

- **120 Days Accrued:** The accrued interest, which is calculated using the coupon payment, business days accrued, and business days in the coupon period, lined to index variation.

- **Total:** The principal plus accrued interest.

- **Number of Certificates:** The total number of notes in the transaction.

- **Real Cash Value in BRL:** The real cash value of the transaction (in Brazilian reals).

- **Accrued Interest:** Accrued interest is calculated using the coupon payment, business days accrued, and business days in the coupon period, linked to index variation.

- **Coupon Calculation:** Coupons are calculated on an exponential basis and inflation-index pro rata. Calculations are according to ANDIMA's standard.

National Treasury Inflation-Linked Bond Series C (NTN-C): NOTA DO TESOURO NACIONAL SERIES C (Calc Type 1195)

Brazil Inflation linked securities, linked to the IBREIGPM Index. Calculations are based on the number of business days in a year being equal to 252 days. These bonds are dirty and issued in unitary price, i.e., quoted in units. Dirty price means that accrued interest is included in the unitary price. The total payment invoice is the unitary price multiplied by the number of bonds. The Par Amount is adjusted by index variation.

- **From Issue to Maturity:** The number of actual days and the number of business days from the issue date to the maturity date.

- **Quote%:** The quote percentage of par value (updated).

- **Price:** The current price of the security, calculated using a discounted cash flow based on the coupon and the yield. The security is linked to the index variation and is traded dirty and in units. Each unit is worth BRL 1,000.00. Dirty price means that accrued interest is included in this unitary price.

- **Par Value (updated):** The par value of the bond (R$1000.00 per note) updated by index variation.

- **IGP-M @ base date:** The index value in the base date.

- **IGP-M:** The index value in the current date.

- **IGP-M Assumption:** The index assumption, published by Associacao Nacional das instituicoes do Mercado Aberto (ANDIMA).

- **Index Variation:** The index variation, which is based on the IGP-M assumption, calculated exponentially, and taking into consideration the current and passed business days, IGP-M base date, and value.

- **Curr. Month Bus Days:** The date range of the current month and the corresponding number of business days passed in the current month.

- **Bus. Days Passed:** The date range of the current month business days passed and the corresponding number of business days passed.

- **Days Accrued:** The accrued interest, which is calculated using the coupon payment, business days accrued, and business days in the coupon period, lined to index variation.

- **Real Cash Value in BRL:** The real cash value of the transaction (in Brazilian reals).

CPI-Based Floaters

This section describes the calculations used in yield analysis for CPI-based floaters.

Consumer Price Index based floaters pay a variable coupon based on changes in the CPI-U index over the period plus one half of the quoted margin, adjusted for the number of days in the period.

The *Yield Analysis* screen for floating rate inflation-indexed bonds is similar to that of the U.S. Treasury's fixed rate inflation-indexed securities (TIPs). Although the two structures may seem different, the yield and sensitivity characteristics of both structures are similar. As with the TIP bonds, there are elements of the *Yield Analysis* screen that quantify the critical characteristics of inflation-indexed securities.

Sensitivity Analysis

Sensitivity numbers are calculated using three different yield beta assumptions for effective duration, risk, and convexity. Whether fixed or floating, duration and convexity measures for both structures are similar. The major difference is the timing of the inflation accrual payments. The TIP bond, the fixed rate structure, pays investors the entire principal inflation accrual at maturity. The floating bond, the floating rate structure, pays the inflation accrual every period.

Note: The sensitivity analysis is highly dependent on the relationship between changes in interest rates and changes in CPI.

- **Yield Beta Assumption:**

  Yield beta, the slope of the regression line, attempts to quantify the change in CPI for a given change in interest rates.

  A yield beta of 0 implies that changes in interest rate and CPI have no relationship (e.g., if interest rates change by 100 basis points, there is no affect on the inflation rate [rate of change of CPI]). Therefore, duration, risk, and convexity measures are similar to that of a fixed rate bullet bond.

  A yield beta of 1 implies that changes in interest rates and CPI move in absolute unison (e.g., a 100 basis point change in interest rates implies, on average, a 100 basis point change in the inflation rate). Therefore, duration, risk, and convexity measures are similar to that of a floating-rate bond.

  Note: The middle column reflects your yield-beta assumption preference from the *Bond Calculation Override* (COVR) function. For more: COVR Help Page.

- **Effective Duration:** The adjusted duration for the projected cash flow.

Danish Mortgage Bonds

This section describes the calculations used in yield analysis for Danish mortgage bonds. YAS calculates the yield (internal rate of return) given a price based on the scheduled payments as published by the Copenhagen Stock Exchange.

Description of Danish Mortgage Bonds

The Danish mortgage bond market consists of two types (Perpetual/Floaters/Index Linked bonds are disregarded in this outline) of bonds:

- Bullet ("plain vanilla") bonds

- Amortizing bonds

The following characteristics apply to both types:

- The conventional yield is always expressed on an annual basis.
- The coupons are usually paid on a quarterly or semi-annual basis, recent issues on a quarterly basis.
- The accrued interest and the price/yield calculation are done on an ACT/ACT day count basis.
- The bonds are traded on a price basis.
- The settlement date = Trade Date + 3 business days.
- The coupon/date/maturity for new bonds issued by the various mortgage institutions are identical.
- The bonds are usually issued as "Tap Issues."
- There is no withholding tax on the bonds in Denmark.
- The turnover of benchmark bonds is very high, therefore liquidity is excellent.
- The bonds go ex-drawing approximately 12 weeks before the coupon date for semi-annual paid bonds and approximately six weeks for quarterly paid bonds.
- The mortgage bond market is larger than the government bond market.
- The bonds do not exist in paper form (only as data entries in the Danish Clearing System).
- Usually the bonds are non-callable (not to be confused with prepayments) and non-puttable.

In respect to bullet issues, Bloomberg treats the bonds like any other straight bond in the system, e.g., Bloomberg automatically generates the cashflows via the information stored on the *Description* (DES) function.

Amortizing bonds may be divided into two categories and each one again into two further groups:

- Annuity Bonds:
  - Prepayments allowed ("Konvertible")
  - Prepayments not allowed ("Non-Konvertible")
- Serial Bond
  - Prepayments allowed ("Konvertible")
  - Prepayments not allowed ("Non-Konvertible")

The yields, sensitivity measurements, etc., are calculated based on the cash flows published by the Copenhagen Stock Exchange.

The cashflows are downloaded daily onto Bloomberg Professional®.

Annuity bonds are repaid by a fixed amount on the specified coupon dates. The amount consists of an interest and an installment portion.

For new issues, the interest portion far outweighs the installment portion, but this ratio changes gradually over the course of the bond's life and reverses in later years. The new issues can, in principle, be compared to standard American mortgage bonds, where the repayment of principal also increases over time.

Serial bonds behave just like any other standard sinking fund bond, as far as repayments are concerned. In other words, the principal is usually repaid by an equal amount over the life of the bond. This means that the total payment decreases with time as the installment portion is fixed and that the interest portion decreases as the outstanding amount decreases.

If a bond is "Konvertible," prepayments are allowed (nothing to do with bonds that are convertible into shares). This means that a holder (investor) of a Danish mortgage bond may be repaid earlier or, rather, with larger amounts than stipulated in the original schedule.

Early repayments usually only take place if the price of the bond moves above 100, so prepayments are not really taken into account in the yield calculations for bonds below par. How much below par is mainly a function of the maturity of the bond and the composite of the debtors for each of the bond issues.

If prepayment is not allowed, the bonds repay according to the schedule, and prepayments can be disregarded in the yield calculations.

Note: The above information is only a brief general outline. For more information, contact your Bloomberg sales representative or press the <Help> key twice. You can also contact the issuer or your Broker/Bank in Denmark.

Prepayment Speed Assumptions

The prepayment speed models are only applicable to Danish Mortgage bonds that are "Konvertible" (i.e., may be prepaid).

CPR Model

The CPR (Constant Prepayment Rate) model assumes that the prepayments are constant for the periods, in percentage. You can enter CPR rates expressed as a period rate or as an annual rate. If you enter a period rate, the annual rate is automatically calculated and vice versa.

For example, a 10% period rate corresponds to an annual rate of 34.4% for quarterly paid bonds. The calculations are:

Start value 100.00

1st Quarter 100.00 - 10% = 90.0

2nd Quarter 90.00 - 10% = 81.0

3rd Quarter 81.00 - 10% = 72.9

4th Quarter 72.90 - 10% = 65.6 = End value

Start value 100.00 vs. End value 65.6 = 34.4% (Annual basis)

Using the same principles, a period rate of 10% corresponds to an annual rate of 19% for a semi-annual paid bond.

The CPR model allows you to control the prepayment amount payable on the first coupon date independent of the remaining dates. You can enter values between 0 and 100%. If you enter 100% in the 1st Period, the assumption is that the whole issue is repaid at the next coupon date.

For example, if you enter 10% for the 1st period and 100% for the remaining, YAS assumes that 10% of the outstanding issue (after deduction of the scheduled payment) is prepaid at the 1st coupon date and the remaining issue is prepaid at the following coupon date.

Note: YAS recognizes if you are in the ex-draw and/or ex-coupon period and adjusts the cash flows to calculate the yields accordingly.

CPR Model Examples and Calculations

The following is an illustration of how the CPR model changes the schedule payments:

Realkredit Danmark 10% 10/01/1997 (quarterly coupon payments)

Cash flows per 5/1/96 are:

Coupon Date	Installment	Interest	Sum
07/01/96	7,052,312	429,580	7,481,892
10/01/96	6,358,002	253,272	6,611,274
01/01/97	2,354,922	94,322	2,449,244
04/01/97	963,360	35,449	998,809
07/01/97	398,159	11,365	409,524
10/01/97	56,450	1,411	57,861

Outstanding amount = 17183205

If prepayment does not take place, the numbers under the *SCHED* and *PREPAY* columns appear as:

Cpn Date	Sched	Prepay
07/96	41.042*	
10/96	37.001**	

Cpn Date	Sched	Prepay
01/97	13.705	
04/97	5.606	
07/97	2.317	
10/97	0.329	

Simply: Installments / Outstanding amount x 100

* 7,052,312 / 17,183,205 x 100 = 41.042%

** 6,358,002 / 17,183,205 x 100 = 37.001%

and so forth.

If CPR is 10% for the 1st period only, the future cashflows change and the numbers under the *SCHED* and *PREPAY* columns appear as:

Cpn Date	Sched	Prepay
07/96	41.042*	5.896**
10/96	33.301***	
01/97	12.334****	
04/97	5.046	
07/97	2.085	
10/97	0.296	

* 7,052,312 / 17,183,205 x 100 = 41.042%

** Outstanding amount per 10/96 is then 10,130,893 and 10% is prepaid of this amount (= 10% CPR for 1st period). The PREPAY column displays (10,130,893 x 10%) / 17,183,205 x 100 = 5.896%

*** (6,358,002 - (6,358,002 x 10%)) / 17,183,205 x 100 = 33.301%

**** (2,354,922 - (2,354,922 x 10%)) / 17,183,205 x 100 = 12.334%

and so forth.

If CPR is 10% for the 1st period and 10% for the remaining periods, the future cash flows change and the numbers under the *SCHED* and *PREPAY* columns appear:

Cpn Date	Sched	Prepay
07/96	41.042*	5.896**
10/96	33.301***	1.976****
01/97	11.101*****	0.668
04/97	4.087	0.193
07/97	1.520	0.022
10/97	0.194	

* 7,052,312 / 17,183,205 x 100 = 41.042%

** Outstanding amount per 10/96 is then 10,130,893 and 10% is prepaid of this amount (= 10% CPR for 1st period). Hence the *PREPAY* column displays (10,130,893 x 10%) / 17183205 x 100 = 5.896%

*** (6,358,002 - (6,358,002 x 10%)) / 17,183,205 x 100 = 33.301%

**** Outstanding amount after payment of 1st scheduled (07/96), 1st prepayment (07/96), and 2nd scheduled reduced by 10% is 17,183,205 - 7,052,312 - (10,130,893 x 10%) - (6,358,002 - (6,358,002 x 10%) = 3,395,602. Since a 10% CPR number is in use in this example for all the remaining payments, the *PREPAY* column displays (3,395,602 x 10%) / 17,183,205 x 100 = 1.976%

***** Scheduled payment was 2,354,922 per 1/1/97. However, on 7/96 and 10/96 prepayment has taken place so the scheduled amount paid is (2,354,922 - 10%)- 10% = 1,907,487. The *SCHED* column displays 1,907,487 / 17,183,205 x 100 = 11.101% and so forth.

SPR Model

The SPR model (Standard Prepayment Rate) is a proprietary model developed by Unibank Markets, Copenhagen, especially for the Danish mortgage bond market. The SPR model assumes a "burn-out" effect so the prepayments decrease over time, which is not the case for the CPR model. The "burn-out" effect occurs at a certain speed (set to 0.24 in the model) so the prepayment after the first year is only 0.24 times the first prepayment.

You can control the first period independently from the remaining payments, as with the CPR model. You can use values between 0 and 100%. For example, if you enter 100% in the first period the assumption is that the whole issue is repaid at the next coupon date. If you enter 10% for the first period and 100% for the remaining periods, the model assumes that 10% of the outstanding issue (after deduction of the scheduled payment) is prepaid at the first coupon date and the remainder is paid on the following coupon date.

Note: YAS recognizes if the settlement date is in the ex-draw period and adjusts the cash flows to calculate the yield accordingly.

If you enter a number between 0 - 99% as the SPR first period rate and/or a number between 10 - 99% as the SPR annual rate, the effect on the cash flows appear in the *SCHED* and *PREPAY* columns.

A period rate is calculated from the annual rate by using the reverse principles explained in the CPR Model.

If you enter a SPR number that matches your estimated prepayment profile, YAS calculates the yields based on the new generated cash flows.

If you enter a number between 0 - 10% as the SPR annual rate, YAS reverts to the CPR model and the numbers move to the CPR fields. This occurs because the model assumes no burn-out effect below a pre-set level, as dictated by the model (set to 10%).

In addition, if the percentage figures (as seen in the *PREPAY* column) produced by the SPR model are less than the corresponding numbers produced by the CPR model, the CPR figures are used for the remaining cash flows. This occurs because of SPR's properties.

When using the models, the figures in the *SCHED* and *PREPAY* columns display the effects on the cash flows and recalculates the *SCHED* & *UNSCHED* yields.

Fixed Income ETFs

This section describes the calculations used in yield analysis for fixed income ETFs.

Fixed Income ETF Calculation Methodologies

For **single currency corporate and government bond ETFs**, an advisory group consisting of Bloomberg, BlackRock, and State Street Global Advisors has developed a market convention, known as the Aggregate Cash Flow method (ACF), for calculating yield, spread, and duration for fixed income ETFs.

The ACF method treats the ETF's holdings as one "big bond" by generating and aggregating cash flows for the individual bond holdings using information from the official ETF provider reports. The ACF method then determines the implied cash for the ETF (the difference between the fund's net asset value and bonds' market value) and treats the implied cash as the first cash flow in the series of aggregated cash flows. The aggregated cash flows are scaled by the ETF shares outstanding times one million to arrive at a "per share" cash flow amount. The aggregated period by period cash flows can be viewed in the *Cashflow Analysis* (CSHF) function.

The advantages of the ACF method include:

- Provides a standard method for valuing Fixed Income ETFs
- Makes it easier to compare FI ETFs with one another and other fixed income instruments, like individual bonds
- Incorporates real-time intra-day underlying bond pricing information into valuation and risk analytics

For more on the ACF method: Aggregate Cash Flow Method for Yield and Duration Calculation for Fixed Income ETFs.

For **all other fixed income ETFs**, YAS uses an internal rate of return (IRR) method for calculating an ETF's yield. Each ETF portfolio is converted nightly into a "Big Bond" by aggregating all individual bonds' cash flows into one large cash flow structure. YAS looks for the interest rate that makes the aggregated discounted cash flows equal to the "Average Bond Price" of the ETF portfolio.

The "Average Bond Price" is calculated by dividing the Asset value of the ETF (number of shares outstanding multiplied by the ETF market price) by the total par value of the bonds after adjusting for cash in the fund.

Inputting ETFs from Other Providers

As long as a provider contributes their ETF's holdings nightly, Bloomberg can model the securities in YAS.

Note: Multi-currency ETFs cannot be modeled for YAS at this time

G-Spread and ETFs

G-spread is the difference in yield of the ETF versus the interpolated yield of the currency matched benchmark curve (US Treasury curve - I25). The benchmark curve is calculated for the interpolated point on the curve that matches the ETF's weighted average maturity (WAM).

I-Spread and ETFs

I-spread is the difference in yield of the ETF versus the interpolated yield of the currency-matched benchmark swap curve (US Dollar Swap curve -30/360 SA). The benchmark curve is calculated for the interpolated point on the curve that matches the ETF's weighted average maturity (WAM).

Floating Rate Debt

This section describes the methodology used to calculate yields on floating rate debt.

To calculate yields on floating rate debt, the previous payment date and all coupon changes since that date must be known, so accrued interest can be calculated. Also, the next coupon fix date must be known.

Yield can be expressed as a margin over the benchmark rate, as in adjusted simple margin, total margin, and discount margin. For example, if a security with five years to maturity floats 25 basis points (bp) over the benchmark rate at a price of $100, then all three margin calculations yield 25bp. If the price is 99.50, then the.50 is divided by five years and 10 is added to the margin over the benchmark. Alternatively, the money market equivalent values the debt as a short term security maturing at the next refix date and calculates a yield.

French Variable Rate Bonds

This section describes the calculations in yield analysis for French variable rate bonds.

Types of French Variable Coupons

There are two types of French variable coupons:

- Pre-fixed: The coupon fixes before the start of the coupon period.
- Post-fixed: The coupon fixes at the end of the coupon period.

Post-fixed coupons are relatively unique to France. The coupon is generally linked to an underlying reference rate such as the TME (Taux Moyen des Emprunt d'Etat). The TME is the average yield for French Government bonds with over seven years remaining life, calculated on a weekly basis.

Crystallization

The principle of crystallization is unique to the French market. This methodology is used to calculate coupons, yields, and margins. In calculating the internal rate of return of the security, all known future cashflows are used and all yet to be determined cashflows are assumed to be fixed (crystallized) at the current market reference rate plus/minus any quoted margin. In addition, certain bonds have coupons that are calculated when the reference rate is multiplied by a given factor. The resulting yield on the bond is compared to the reference rate to see whether it is trading over or under that rate.

In the case of pre-fixed bonds, the crystallization only applies to future coupons. For post-fixed bonds, it is also used in the process to estimate an element of the current period's coupon, as well as future coupons.

Italian Governments

This section describes the calculation of Italian government bond conventional yields post reconvention.

- **BOTS (Bills):** Bills issued before Jan. 1, 1999 use simple "true" gross yield (maturity date adjusted for weekends using the Italian calendar) and an act/365 day count.

  Bills issued after Jan. 1, 1999 use act/360 day count for new issues. Uses simple "true" gross yield (maturity date adjusted for weekends using the Trans-European Automated Real-Time Gross Settlement Express Transfer system calendar) and act/360 day count.

- **ICTZ (Zero coupon bonds):** Bonds maturing before Dec. 31, 1999 use simple "true" gross yield (maturity date adjusted for weekends using the Italian calendar) and act/365 day count.

  Bonds maturing after Jan. 1, 2000 use annual compounded "true" gross yield (maturity date adjusted for weekends using the Target calendar) and act/act day count. For issues with a remaining life of one year or less, simple yield is applied.

- **BTPSS/BTPSR (Strips):** Uses annual compounded "true" gross yield (maturity date adjusted for weekends using the target calendar) and act/act day count. For issues with a remaining life of one year or less, simple yield is applied. The calculation pre Jan. 1,1999 uses the Italian calendar for adjustment of the maturity date, if applicable, and act/365 day count.

- **BTPS & CTES (Government Bonds):** For Italian Governments, if the first coupon date in 1999 falls on Jan. 1, 1999, then reconvention takes place on the second coupon payment date in 1999.

  The calculation of the accrued interest for issues reconventioned is based on an act/act day count and calculated as:

  ((c * 100)/n) * (Dt/Ds) = ai (round to 5 decimals). Then, ai / 100 * face = amount to calculate the total accrued interest amount for the position, where:

  ai = accrued interest (per nominal 100)

  c = annual coupon

  n = number of coupons paid in the year

  Dt = actual number of calendar days between the settlement date and the previous coupon date

Ds = actual number of calendar days in the current interest period, i.e., from last coupon date to the next coupon date

The price to yield calculation is also based on an act/act day count basis and the conventional "true" gross yield (coupons adjusted for weekends using the target calendar) always appears on an annual compounded basis, regardless of the frequency of the coupon payments.

For bonds not yet reconventioned, the calculation is based on a day count of 30/360 +1 for the accrued interest and a day count of act/365 for the price to yield calculation. The accrued interest is calculated as:

(c * 100) * ((no. of days from last coupon date to settl. date / 360) + 1) = ai (round to 5 decimals). Then, ai / 100 * face amount = the total accrued interest amount for the position.

However, the price to yield calculation is based on an act/365 day count and the conventional "true" gross yield (coupons adjusted for weekends using the Italian calendar) always appears on an annual compounded basis.

- CCTS (Floating Rate Notes): For Italian Governments, the discount margin and the accrued interest calculation for issues reconventioned uses an act/act day count. For issues not reconventioned, the discount margin used 30/360 day count. Reconvention rules and accrued interest calculation are the same as BTPS. For more information, see the BTPS entry above.

Non-U.S. Inflation-Indexed Bonds

This section describes the calculations used in yield analysis for non-U.S. inflation-indexed bonds.

Sensitivity Analysis and Yield-Beta Assumptions

To better assess the risk of inflation protected securities, YAS's sensitivity analysis appears for three different assumptions of yield-beta. Yield-beta attempts to measure the slope of the relationship between nominal yields and real yields. To enter personal defaults for yield betas, use the *Bond Calculation Override* (COVR) function. For more: COVR Help Page.

A yield-beta assumption of one means that there is a perfect correlation between the movement of nominal rates and real rates. In this case, sensitivity analysis is performed (as with normal bullet bonds) by changing the yield slightly and measuring the effect on price, all while maintaining the constant inflation assumption shown in the *Inflation Assumption* field. Hence, a one basis point shift in nominal rates causes the same one basis point shift in real rates. The risk values under this scenario are similar to that of nominal bonds.

A yield-beta assumption of zero means that there is no correlation between the movement of nominal rates and the movement of real rates. Under this assumption, for a given change in nominal rates, there is no change in real rates. This causes the risk values to be quite low.

A yield-beta value of 0.5 is an arbitrarily selected value to help illustrate the effect of different beta assumptions. Under this scenario, a one basis point shift in nominal rates causes a half basis point change in real rates.

Economic Factors

Economic Factors are used to quantify the rate of inflation, so inflation-adjusted coupon payments and the inflation-adjusted principal can be calculated.

The principal value of the notes changes based on the frequency of the CPI index. Daily accretion is based on straight-line interpolation, using the actual number of days in each month. There is a three-month lag, so, for example, the June 2000 CPI is the index level used for Sept. 1, 2000. The CPI is then turned into a ratio to calculate the accretion factor, by taking the index on the date and dividing by the Base CPI Value. The following factors are used in the calculations:

- **Base CPI Value (Issue Date):** The base CPI value at the security's issue date.
- **Reference CPI (Settlement Date):** The adjusted CPI value as of the settlement date.
- **{ticker symbol} <INDEX> {index date}:** The first non-seasonally adjusted CPI index divided by the Base CPI Value. To display the single-security functions menu for the index, enter {ticker symbol} <INDEX> <GO>.
- **{ticker symbol} <INDEX> {index date}:** The second non-seasonally adjusted CPI index divided by the Base CPI Value. To display the single-security functions menu for the index, enter {ticker symbol} <INDEX> <GO>.
- **CPI @ Last Cpn Date:** The consumer price index value at the selected security's last coupon date.
- **Flat Index Ratio:** The increase or decrease in the inflation rate between the issue date and the last coupon date, calculated as CPI @ Last Coupon Date/Base CPI Value.
- **Accrued Ratio Growth:** The rate of inflation from the last coupon date to the settlement date, calculated as *Index Ratio - Flat Index Ratio*.
- **Index Ratio:** The increase or decrease in the inflation rate between the issuance date and the settlement date, calculated as Reference CPI/Base CPI Value.

Note: The Index Ratio is essential in valuing the security, because it is the factor that is used to adjust the accrued interest and principal amount to properly calculate principal and interest at settlement. The three ratio numbers - Flat Index Ratio, Accrued Ratio Growth, and Index Ratio - quantify how the inflation rate has changed from the issue date and from the last coupon date.

Calculations

To find the CPI for any date in January, take the difference between the two index levels, divide by the actual number of days in the month, multiply the result by the date - 1 (January 7 would be 6), and add the result to the January 1 value. For example, with a January 1 level of 158.30 and a February 1 level of 158.60:

158.60 - 158.30 =.30

.30/31 =.0096774

.0096774 * 6 =.05806

CPI for Jan. 7 = 158.30 +.05806 = 158.35806.

The CPI for January 15 (the official issue date of the bonds) is 158.43548.

The CPI for January 25 is 158.53226. The index ratio for January 25 is 158.53226/158.43548 = 1.00061.

If the coupon is set at 3.5%, and the index ratio is 1.01 at the coupon date, the actual coupon payment is (100 * 1.01 * 0.035)/2 = 1.7675 per 100 bonds.

Pay-in-Kind Bonds

This section describes the calculations used in yield analysis for pay-in-kind bonds.

PIK bonds do not pay interest. Instead, the issuer pays the holder in additional bonds. This occurs until some future date, when the bond may convert into a normal interest paying security.

Bloomberg values PIKs by accumulating the additional bonds and calculating their restated principal value by compounding them at their coupon rate. For example, a semi-annual 16% PIK bond increases in face value by 8% compounded semi-annually, until it converts into an interest paying bond. The increase in redemption value appears in the *Pay-In-Kind Redemption Value Growth Factor* field.

South African Government Bonds

This section describes the calculations used in yield analysis for South African Government bonds.

- The bonds are traded on a yield basis.
- The settlement date is T+3.
- The accrued interest is based on a day count of ACT/365, and the bonds go ex-coupon (negative accrued interest) 10 calendar days before the coupon date.
- The price to yield calculation (the internal rate of return) uses ACT/ACT day count, and the yield is shown on a semi-annual compounded basis. However, in the final coupon period, simple yield is used.
- The gross (dirty) price and the capital (clean) price are always shown with decimals per nominal 100. The proceeds are rounded to two decimals.

The conventional rounding of the dirty price, accrued interest, and clean price follow these steps:

1. Given a yield, the net present value (gross price) per nominal 100 is calculated (un-rounded).

2. Then the accrued interest per nominal 100 is calculated (un-rounded).

3. Deduct the accrued interest from the net present value to obtain the capital (clean) price (un-rounded).

4. Round the capital price and the accrued interest to five decimals.

5. Add the rounded capital price to the rounded accrued interest to reconstruct the gross price (which appears with five decimals).

South Korean Domestic Bonds

This section describes the calculations used in yield analysis for South Korean domestic bonds.

South Korean bond yield calculations are based on the following data:

P = The bond price

Sn = The total principal and interest (redemption amount at maturity)

r = The trading yields per year

d = The remaining number of days from the settlement date to the interest payment date

cr = The coupon rate

m = The number of interest payments per year

n = The number of years to the expiration date

In = Interest

Habitual Yield Calculation

- **Discount Bonds: (calc. type 776) - Monetary Stabilization Bonds, Industrial Finance Bonds:**

  $P = Sn / ((1+r)^{(n-1)} (1+r \times d/365))$

  (Sn = Face Value; 10,000 Won)

  Where,

  $(1+r \times d/365) >= (1+r)^{(d/365)}$ thus, $1/(1+r \times d/365) <= 1/((1+r)^{(d/365)})$

- **Simple Interest Bonds (calc. type 776) - (like discount bonds, with simple interest at maturity):**

  $P = Sn / ((1+r)^{(n-1)} (1+r \times d/365))$

  $(Sn = 10{,}000 \times (1 + cr \times n))$

- **Compound Bonds (calc. type 764) - quarterly compounded:**

  - Quarterly Compounded

    $P = Sn / (1+r)^{(n-1)} (1+r \times d/365)$

    $(Sn = 10{,}000 \times (1 + cr/4)^{(4n)})$

  - Annually Compounded

    $P = Sn / (1+r)^{(n-1)} (1+r \times d/365)$

    $(Sn = 10{,}000 \times (1+ cr)^n)$

- **Coupon Bonds (calc. type 768) - Corporate Bonds, Special Public Bonds:**

  $P = In / (1 + r/m)^0 + \dots + In / (1 + r/m)^{(n-2)} + (10{,}000 + In) / (1 + r/m)^{(n-1)}$ over $(1 + r/m)$ * (No. of remaining days of a payment period / Total No. of days of a payment period)

Korean bond interest on income tax calculations are specific to each bond type.

Discount and Simple Interest Bonds

- **Interest on Income:**

Face Amount x discount rate x holding period/365

(fractions under 1 Won deleted).

- **Holding period:** The purchase date (included) to the selling date (or redemption date, disregarded). Leap month is included in case the holding period is less than one year.
- **Leap year:** Disregarded (a year is always 365 days).

Note: Interest tax is the tax rate percentage of Interest on Income. Resident tax is the tax rate percentage of the interest tax amount.

Compound Bonds

Interest on Income* = Face Amount x [(1+r/n)^(c+d/t) - (1+r/n)^(m+e/t)], where:

r = The issue rate

n = The number of payment dates per year

c = The number of n-periods between the issue date and selling date

d = The number of days to the selling date after the last payment date (not included), including leap year

m = The number of n-periods between the issue date and purchase date

e = The number of days to the purchase date after the last payment date (not included), including leap year

t = The number of days during the payment period, including leap year

Note: Interest tax is the tax rate percentage of Interest on Income. Resident tax is the tax rate percentage of the interest tax amount.

Coupon Bonds

- **Interest on Income:** (Face Amount x r/n x holding period (1) / number of days during payment period) + (Face Amount x discount rate x holding period (2) / total number of days to expiration date).
- **Holding period (1):** The number of days between the purchase date (or previous payment date) and the selling date (or next payment date or expiration date), including leap year.

  Holding period (1) is comprised of the number of days in the payment period, including leap year or the total number of days to expiration date, including leap year.
- **Holding period (2):** The number of days between the purchase date and the selling date (or expiration date), including leap year.

Note: Interest tax is the tax rate percentage of Interest on Income. Resident tax is the tax rate percentage of the interest tax amount.

U.S. TIPs

This section describes the calculations used in yield analysis for U.S. TIPs.

Economic Factors are used to quantify the rate of inflation, so the inflation-adjusted coupon payments and the inflation-adjusted principal can be calculated.

The principal value of the notes changes based on the frequency of the CPI index. Daily accretion is based on straight-line interpolation, using the actual number of days in each month. There is a three-month lag, so, for example, the June 2000 CPI is the index level used for Sept. 1, 2000. The CPI is then turned into a ratio to calculate the accretion factor by taking the index on the date and dividing by the Base CPI Value. The following factors are used in the calculations:

- **Base CPI Value (Issue Date):** The base CPI value at the security's issue date.
- **Reference CPI (Settlement Date):** The adjusted CPI value as of the settlement date.
- **{ticker symbol} <INDEX> {index date}:** The first non-seasonally adjusted CPI index divided by the Base CPI Value. To display the single-security functions menu for the index, enter {ticker symbol} <INDEX> <GO>.
- **{ticker symbol} <INDEX> {index date}:** The second non-seasonally adjusted CPI index divided by the Base CPI Value. To display the single-security functions menu for the index, enter {ticker symbol} <INDEX> <GO>.
- **CPI @ Last Cpn Date:** The consumer price index value at the selected security's last coupon date.
- **Flat Index Ratio:** The increase or decrease in the inflation rate between the issue date and the last coupon date, calculated as CPI @ Last Coupon Date/Base CPI Value.
- **Accrued Ratio Growth:** The rate of inflation from the last coupon date to the settlement date, calculated as INDEX RATIO - FLAT INDEX RATIO.
- **Index Ratio:** The increase or decrease in the inflation rate between the issuance date and the settlement date, calculated as Reference CPI/Base CPI Value.

Note: The Index Ratio is essential in valuing the security, because it is the factor that is used to adjust the accrued interest and principal amount to properly calculate principal and interest at settlement. The three ratio numbers - Flat Index Ratio, Accrued Ratio Growth, and Index Ratio - quantify how the inflation rate has changed from the issue date and from the last coupon date.

Venezuelan DPNs

This section describes the calculations used in yield analysis for Venezuelan DPNs.

- **Yield:** The annual rate of return based on the price, using an iteration methodology, and calculated as:

  $Y = ((1 + [C\%*n/360]) / Px\%) * (360/n * 100)$

  where:

  Y = yield

  $C\%$ = current coupon

n = number of actual days from the issue/last

coupon date to maturity

Px = price

- **Repo to (next refix date):** The next re-fix rate and assumed finance rate for the time period from settlement to the next re-fix date. This rate is only used as the financing rate in the adjusted price calculation. For DPNs, the other floating coupons are calculated as:

  CF = RP% * TAM(5d, prev cpn date)

  where:

  CF = next floating coupon

  RP = reference percentage

  TAM(5d, prev cpn date) = the TAM released five days prior to the last coupon period

  Note: The Central Bank of Venezuela usually releases the TAM rate with a one-week delay. Even if the rule to calculate the next coupon says to use the TAM rate five business days prior to the last coupon date, it is necessary, in practice to use the rate 10 days prior to the last coupon. Before the Central Bank releases the rate, Bloomberg estimates the next coupon using the most recent rate released.

- **Days Accrued Interest:** The amount of interest accumulated, but not paid, between the issue date or most recent payment date and the settlement date.

  Note: For bonds issued prior to July 27, 2000, the day type is 30/360+1. For bonds issued between July 27, 2000 and February 20, 2001, the day type is 30/360 and for bonds issued after February 20, 2001, the day type is ACT/360, considering a period of 91 days per quarterly coupon period.

Warrants

This section describes the calculations used in yield analysis for warrants.

For more on warrants: Definitions.

Warrant Classes

Bloomberg recognizes two broad classes of warrants, which are based on how the warrant is priced:

- **Percent Type:** The warrant price is expressed as a percent of the par amount of the underlying bond (CALCTYPE = 126). Japanese yen equity warrants typically are percent types.

- **Fixed Type:** The stated warrant price is the value (CALCTYPE = 125). Swiss franc equity warrants are fixed types.

For more on calculation types: Brochures.

Warrant Horizon Analysis

YAS performs an analysis based on the stock price that you select and four horizon scenarios, numbered I to IV. The current set of values details the analysis for the latest set of values. The starting values of the horizon scenarios are determined from the current set of values as:

	I	II	III	IV
Stock Price (SP = Current Stock Price)	90%*SP	95%*SP	105%*SP	110%*SP
Percent Premium (Current Value +/− {X})	+4	+2	−2	−4
Exchange Rate	Current	Current	Current	Current

The output values for these scenarios are the warrant parity, warrant price, and gearing.

Note: All prices are in the same currency. Use the exchange rate to convert the warrant price into the currency of the stock.

General Calculations

This section includes calculations that are not specific to a particular bond subtype.

Modified Duration

Modified duration is calculated as:

[duration / {1 + (IRR/M)}]

where

IRR = the internal rate of return and M is the number of compounding periods per year.

Simple Yield Conversion Method

Simple yield is converted as follows:

[1 + Y1 / (100*F1)]^F1 = [1 + Y2 / (100*F2)]^F2

where

Y1 = bond's conventional yield

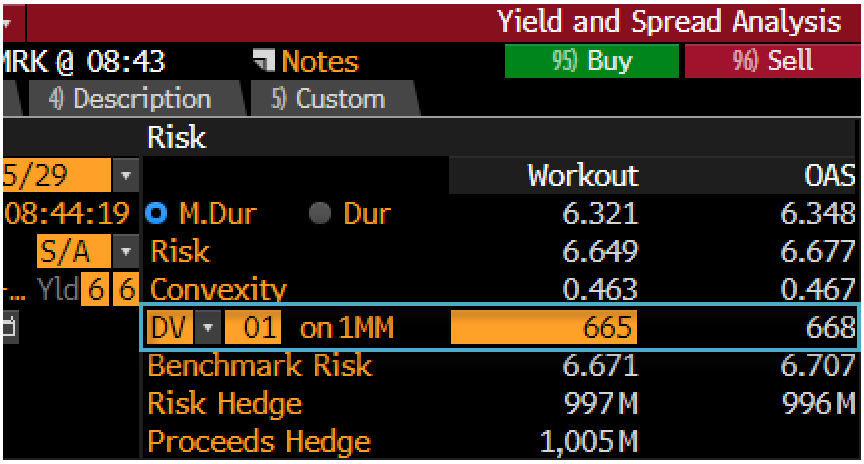
Y2 = equivalent yield

F1 = bond's coupon payment frequency

F2 = equivalent yield's frequency of coupon payment (= xx)

DV01 / PV01 / YV01

In the *Risk* section of the *Yield & Spread* tab, you can calculate the dollar value (DV) or price value (PV) of a shift in rates for a bond, or the yield value (YV) of a shift in price for the bond.



You can specify the amount of the shift to apply. For example, in the image above, YAS is calculation the dollar value of a 1 bps shift in rates.

- **DV 01:** The dollar value change in market value given a one basis point change in interest rates. It is calculated as Risk /100, when Risk = dirty price * Mod duration/100
- **PV 0.01:** The dollar amount by which the market value of the bond changes if the coupon changes by one basis point (0.01%).
- **YV 0.01:** The value by which the yield of the bond changes if the price changes by one basis point (0.01%).

Shortcuts

YAS consolidates the *Yield Analysis* (YA), *YAS for Bond Quote* (YASQ), *Yield to Call Analysis* (YTC), and *Yield to Put Analysis* (YTP) functions into one function. When you enter any of these mnemonics, the consolidated YAS appears.

Accessing

To access YAS, enter (ticker symbol) (coupon) (maturity date) <yellow key> YAS <GO>.

For example:

- F 5.75 02/01/21 <CORP> YAS <GO>
- T 3 ⅛ 05/15/21 <GOVT> YAS <GO>

The *Yield and Spread Analysis* screen appears.

Shortcuts

You can use shortcuts to access specific YAS functionality.

Enter (ticker symbol w/ coupon and maturity) <yellow key>, followed by	To specify
YAS P (price) <GO>	A price.
YAS Y (yield) <GO>	A yield.
YAS S (spread) <GO>	A spread.
YAS SD (mm/dd/yyyy) <GO>	A settlement date.
YAS TD (mm/dd/yyyy) <GO>	A trade date.
YAS DM (discount margin) <GO>	A discount margin.
YAS *(spread) (search code) <GO>	The spread and search convention from the underlying benchmark yield curve. Choose from the following search codes: • [D] Spread to option–adjusted–spread (OAS) duration–matched benchmark. • [M] Spread to the maturity–matched benchmark. • [W] Spread to the benchmark based on the bond's yield–to–worst date. • [C] Spread to next call. • [P] Spread to next put. • [I] Interpolated spread to the curve. • [xM] Where [x] is the maturity month of a bill on the benchmark yield curve (i.e., 6M is the six–month bill).

Enter (ticker symbol w/ coupon and maturity) <yellow key>, followed by	To specify
YAS (override code) <GO>	The override default benchmark. Choose from the following override codes: • [B] The At–Issue benchmark, if available from the bond's description (DES) page. • [M] Spread to the maturity–matched benchmark. • [W] Spread to the benchmark based on the bonds yield–to–worst date. • [xM] Where [x] is the maturity month of a bill on the benchmark yield curve (i.e., 6M is the six–month bill) • [xY] Where [x] is the maturity year of a note or bond on the benchmark yield curve (i.e., 10Y is the 10–year note). • [CTxn] For U.S. securities only, where [x] denotes a maturity (2–30). [n] denotes the off–the–run (A–Z, where A is the on–the–run, B the most recent off–the–run, etc.) • [SB] For Trading Systems clients, to override the trading system benchmark.
YAS SPEX (BBID) (bps) BP	The benchmark issue and the spread to the specified benchmark in basis points. You must use the Bloomberg ID when specifying the benchmark issue.

Excel Integration

You can transfer data from YAS to a Microsoft® Excel spreadsheet by creating API formulas.

API Formulas

Formula Construction

You can create custom API formulas within Microsoft® Excel, so you can analyze one or more securities at once in a spreadsheet as you would on YAS. Custom formulas also allow you to tailor the data to only the most relevant information for

your analysis.
YAS uses BDP and BDP override formulas to import data into Excel.

- For more on constructing BDP, BDP override, and other API formulas: DAPI Help Page, Bloomberg API - Tutorial with Examples.
- For more on searching for data fields to use in formulas: Finding Data Fields.
- For examples of formulas you can create: Example: Yield, Example: Price to Yield, Example: Yield to Price, Example: Modified Duration, Example: I-Spread.

Finding Data Fields

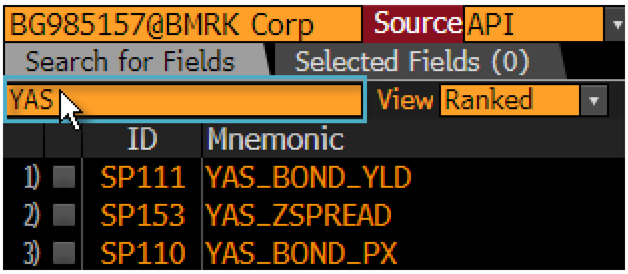
You can find data fields in the *Field Search* (FLDS) function that you can use in API formulas.

For more: FLDS Help Page.

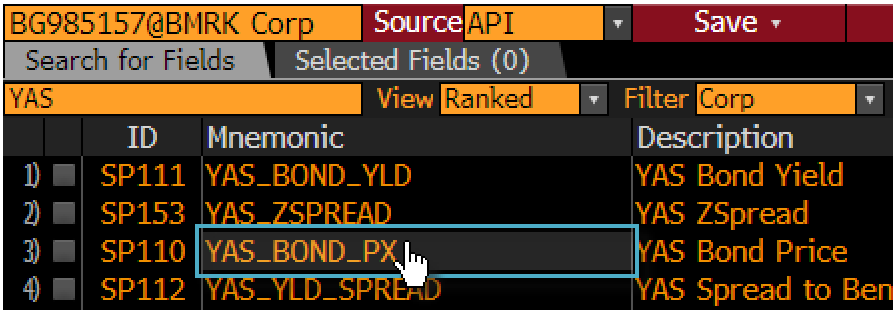
Steps:

1. In the command line, enter a bond ticker (e.g., VZ 3 03/22/27 <Corp>), then press <GO>.
2. In the command line, enter **FLDS**, then press <GO>.

   FLDS appears.
3. In the *Enter Query* field at the top left of the screen, type "YAS," then press <GO>.

   Fields that can be used to download data from YAS appear in the search results. You can use any field that displays a value in the *Value* column.
4. In the search results, click a field (e.g., *YAS_BOND_PX*).



The *Field Information* screen appears for the selected field.

Note: Some field definitions also include a list of override fields that allow you to recalculate the field's value. In the middle of the *Field Information* screen, you can select **Override** to see the corresponding override fields. The field and its override fields can be used in BDP override formulas For more: Example: Price to Yield, Example: Yield to Price, Example: Modified Duration, Example: I-Spread.

Example: Yield

This topic provides a practical example for creating a formula to import the yield of a bond into Microsoft® Excel. This example shows how to import the yield of VZ 3 03/22/27 <Corp>.

Steps:

1. In the command line, enter **VZ 3 03/22/27 <Corp> FLDS** <GO>.

   FLDS appears with your bond loaded.

2. In the *Enter Query* field, type "YAS yield," then press <GO>.

   The YAS_BOND_YLD field appears in the search results.

3. In a cell on your spreadsheet, use the field to create a BDP formula:

   =BDP("VZ 3 03/22/27 Corp","YAS_BOND_YLD")

The yield appears in your spreadsheet.

Example: Price to Yield

This topic provides a practical example for creating a formula to import price to yield data for a bond into Microsoft® Excel. This example shows how to import the yield of BACR 7.625 11/21/22 <Corp>, based on a price of 102 and a settle date of July 29, 2020.

Steps:

1. In the command line, enter **BACR 7.625 11/21/22 <Corp> FLDS** <GO>.

FLDS appears with your bond loaded.

2. In the *Enter Query* field, type "YAS yield," then press <GO>.

   The YAS_BOND_YLD field appears in the search results.

3. To display the field's definition, click the field.

   The *Field Information* screen appears with YAS_BOND_PX and SETTLE_DT listed as overrides.

4. In a cell on your spreadsheet, use the fields to create a BDP override formula:

   =BDP("BACR 7.625 11/21/22 Corp","YAS BOND YLD","YAS BOND PX=102","SETTLE DT=20200729")

The yield appears in your spreadsheet.

Example: Yield to Price

This topic provides a practical example for creating a formula to import yield to price data for a bond into Microsoft® Excel. This example shows how to import the yield of JPM 6 12/29/49 <Corp>, based on a yield of 4.5 and a settle date of July 29, 2020.

Steps:

1. In the command line, enter **JPM 6 12/29/49 <Corp> FLDS** <GO>.

   FLDS appears with your bond loaded.

2. In the *Enter Query* field, type "YAS price," then press <GO>.

   The YAS_BOND_PX field appears in the search results.

3. To display the field's definition, click the field.

   The *Field Information* screen appears with YAS_BOND_YLD and SETTLE_DT listed as overrides.

4. In a cell on your spreadsheet, use the fields to create a BDP override formula:

   =BDP("JPM 6 12/29/49 Corp","YAS BOND PX","YAS BOND YLD=4.5","SETTLE DT=20200729")

The yield appears in your spreadsheet.

Example: Modified Duration

This topic provides a practical example for creating a formula to import modified duration data for a bond into Microsoft® Excel. This example shows how to import the modified duration of 03512TAD3 <Corp>, based on a price of 110 and a settle

date of June 1, 2020.
Steps:

1. In the command line, enter **03512TAD3 <Corp> FLDS** <GO>.

   FLDS appears with your bond loaded.

2. In the *Enter Query* field, type "modified duration," then press <GO>.

   The YAS_MOD_DUR field appears in the search results.

3. To display the field's definition, click the field.

   The *Field Information* screen appears with YAS_BOND_PX and SETTLE_DT listed as overrides.

4. In a cell on your spreadsheet, use the fields to create a BDP override formula:

   =BDP("03512TAD3 Corp","YAS_MOD_DUR", "YAS_BOND_PX=110", "SETTLE_DT=20200601")

The modified duration appears in your spreadsheet.

Example: I-Spread

This topic provides a practical example for creating a formula to import the I-Spread of a bond into Microsoft® Excel. This example shows how to import the I-Spread of US48126HAA86 <Corp>, based on a price of 120 and a settle date of June 30, 2020.

Steps:

1. In the command line, enter **US48126HAA86 <Corp> FLDS** <GO>.

   FLDS appears with your bond loaded.

2. In the *Enter Query* field, type "Ispread," then press <GO>.

   The YAS_ISPREAD field appears in the search results.

3. To display the field's definition, click the field.

   The *Field Information* screen appears with YAS_BOND_PX and SETTLE_DT listed as overrides.

4. In a cell on your spreadsheet, use the fields to create a BDP override formula:

   =BDP("US48126HAA86 Corp","YAS_ISPREAD","YAS_BOND_PX=120", "SETTLE_DT=20200630")

The I-Spread appears in your spreadsheet.

Example: Total Return

This topic provides a practical example for creating a formula to import total return data for a bond into Microsoft® Excel.

Steps:

1. Open the Fixed Income Total Return Calculator spreadsheet, then, in *Securities* section, add the security's Ticker, ISIN, or CUSIP.

2. Specify the start of your holding period. This can be a relative period such as "-X", or a specific date. =BDP([Ticker],[CUST_TRR_START_DT],[STARTDATE]

   - 1D is 1 day prior
   - 3W is 3 weeks prior
   - 4M is 4 months prior
   - 2Y is 2 years prior

3. Specify your reinvestment rate for coupons. You can choose between two options: NONE or FIXED. If you choose FIXED, enter your fixed rate. You can also override using [CUST_TRR_REINVESTMENT_TYP_OVRD],[NONE/FIXED]).

4. Specify the end of your holding period. This can be a relative period, or a specific date, on a security level. [CUST_TRR_END_DT],[ENDDATE]

5. Specify your prices. The sheet will default to the last price as of the specified date. You can override your own custom price used for the calculation in the Price field, using [CUSTOM_TOTAL_RETURN_END_PRICE],[PRICE]) and [CUSTOM_TOTAL_RETURN_START_PRICE](optional),[PRICE].

Brochures

The following documents provide additional insight into YAS and related functionality.

Type	Title	Description
	White Paper: Aggregate Cash Flow Method for Yield and Duration Calculation for Fixed Income ETFs	Details of the Aggregate Cash Flow (ACF) method created by an advisory group consisting of Bloomberg, BlackRock, and State Street Global Advisors and to be applied to single currency government and corporate fixed income ETFs.
	Brochure: Bloomberg for Investment Advisors	An introduction to Bloomberg's news, research, data, and analytics that can help you build assets, management investment performance, and service clients.

Definitions

Term	Definition
(# of days) Days Accrued Int	The amount of interest accumulated, but not paid, between the issue date or most recent payment and the settlement date. The number of days in the accrual period appears to the left of the field.
(Risk Convention) NV Duration	Also known as Macauley's duration. The weighted average maturity of the security's cash flows, where the present values of the cash flows serve as the weights. The greater the duration of a security, the greater its percentage price volatility. The duration of zero coupon bonds is the maturity itself. For all other bonds, the duration is less than the maturity.
(workout date)@(price)	The price at the workout date.
@REDM	The redemption price of the preferred security.
1st Cmpnd	For multi–coupon bonds, the first coupon payment date.
Accrued	The amount of interest accumulated, but not paid, between the issue date or most recent payment and the settlement date. The number of days in the accrual period appears to the right of the field.
Accrued Income	For Australian floating rate notes, the net present value of all accrued interest on the bond.
Accrued Int (XX) Days	For CPI–based floaters, accrued interest is calculated using a straight line accretion between the base CPI–U value and the latest value. **Note:** The cumulative accrued interest may not equal the coupon, as the CPI-U level can suddenly change in the final period that determines the actual coupon.
Accrued Interest	The amount of interest accumulated, but not paid, between the issue date of most recent payment and the settlement date. Bonds in default are analyzed as if the bond is not in default, including the amount of accrued interest that is normally due.
Accrued Interest/Bond	The amount of interest calculated from the previous coupon payment date up to, but not including, the settlement date.
Activity Log Entries	A setting that allows you to set the number of days after which activity log entries are distilled or deleted from the *YAS Activity Panel* (YASA). For more: Displaying YAS Activity.
Adj/Mod Duration	The adjusted or modified duration, which is the percentage price change of a security for a given change in yield. The higher the modified duration of a security, the higher its risk. Adj/Mod Duration = [duration / {1+(IRR/M)}]; where IRR is the internal rate of return and M is the number of compounding periods per year. Duration also mathematically quantifies the price/yield sensitivity of a bond. For example, if one bond has twice the duration of another bond, its relative price change is twice as large for an equal yield change. This is solely a mathematical property for equal yield changes. In the actual marketplace, the observed day–to–day volatility may not be 2–to–1, thereby reflecting unequal non–parallel yield shifts. In its price/yield sensitivity role, duration joins many other related measures, such as price value of a.01, basis point value, yield value of a 32nd, and Bloomberg's *Risk*.

Term	Definition
Adjusted Duration	The percentage price change of a security for a given change in yield. The higher the modified duration of a security, the higher its risk. Ad/ModDuration = [duration / {1 + (IRR/M)}]; where IRR is the internal rate return and M is the number of compounding periods per year.
Adjusted Price	For floating rate notes, the breakeven price, adjusted for the cost-of-carry for settlement at the next refix date. This calculation allows a trader settling a trade at the next refix date to price the security, so it can be financed by a repurchase agreement until the next refix date without generating a loss for the trader. **Note:** This calculation employs the repo to the Next Refix Date as the financing rate, just as the *Cost of Carry* (COC) function uses a repo rate to perform breakeven cost-of-carry analysis for fixed-rate bonds. For more: COC Help Page.
Adjusted Simple Margin	The adjusted simple margin (ASM) contains two return components: the quoted spread over the benchmark, plus the difference between par and the adjusted price, both amortized over the remaining life. The ASM then consists of a locked-in spread over the benchmark, plus (or minus) the bonus (penalty) associated with the positive (negative) carry over the base rate, during the coupon period. For floating rate notes, the ASM represents one of three ways of evaluating margins. This is the simplest method used to evaluate new issues that have not yet received their first coupon, and secondary floating rate notes that are trading flat of accrued interest.
Adjusted Total Margin	The adjusted simple margin (ASM) plus the interest earned by investing the difference between par and the adjusted price (or when adjusted price > 100 minus the interest foregone). The assumed index is used as the investment rate. For floating rate notes, adjusted total margin represents one of three ways of evaluating margins. This method is an attempt to improve and expand upon the ASM method.
Advantage	For convertible bonds, the difference between the bond's current yield and the underlying stock's current yield.
After Tax	The after-tax yield, obtained when coupon income is taxed at the income tax rate and capital gains are taxed at the capital gains rate.
After Tax Yield	For U.K. index linked stock, the after-tax yield, obtained when coupon income is taxed at the income tax rate and capital gains are taxed at the capital gains rate.
AL/Avg Life	Appears when available. The average number of years that each dollar of unpaid principal due remains outstanding.
All Cash Pay Yield	For pay-in-kind bonds and toggle notes, the yield assuming the holder receives a cash payment for all coupons due.
All Cash Yield	For bonds in default, the total yield value. The bond is analyzed as if it is not in default, including the amount of accrued interest that is normally due.
All In	The all in credit spread calculated for the loan, which measures the annualized return on the loan, taking the combined value of the current margin, pay in kind margin(where applicable), and index floor over the life of a loan, as of the settlement date. The Average Life is capped at 4. The all in credit spread does not account for margin ratchets.

Term	Definition
All PIK Yield	For toggle notes, the yield assuming that the bond will pay PIK for life and that the holder will keep the additional bonds they receive, resulting in the holding increasing by the redemption value growth factor.
ALLQ - All Quotes	A component on the *Custom* tab that allows you to monitor current market prices for the selected security using the *All Quotes* (ALLQ) function. The *ALLQ - All Quotes (medium)* option provides a larger panel with more quote information. For more: ALLQ Help Page.
Amt	The value of the corresponding security for finance, expressed in millions.
Annual Dividend	For convertible bonds, the current annual dividend amount for the underlying stock.
Annual Percentage Yield (APY)	An annually compounded value, using an ACT/360 or ACT/365 day type.
Annualized Projected Cpn	The projected coupon on an annual basis.
AOAS - Agency OAS	A component on the *Custom* tab that evaluates callable benchmark agency bonds using Bloomberg option–adjusted spread methodology and market standard defaults, so you can determine how the security reacts to interest rate shifts and compare your OAS analysis to a traditional yield analysis.
Approx Record Date	For pay–in–kind–bonds, the date an investor must be registered with the issuing company as owner of the security to receive payments.
Ask Px	The ask price of the security.
Ask Sz	The amount of the security the market maker is willing to sell at the ask price, where M = thousands and MM = millions.
Asset Swap Calc (Underlying Curves)	The type of price(s) for the underlying swap curve(s).
Assumed Annual Inflation Rate(%)	For U.K. index–linked stock, the rate at which future cash flows are adjusted to compensate for inflation.
Assumed Coupon	For floating rate notes, the assumed fixed coupon for future cash flows.
Assumed Cpns Guaranteed	For Brady bonds, the assumed coupons guaranteed are expressed as a percent of the face value of the bond. You can also specify future dates and rates. For fixed rate bonds, this percentage represents a fixed set of interest payments. For floating rate notes, this can represent additional or fewer coupons, due to the floating nature of the bond.
Assumed Index	For floating rate Brady bonds, the current rate of the index to which the security is linked.
Assumed Rate	For floating rate notes, the assumed average rate of the benchmark for all future refixes, including refixes that may occur before the next coupon payment date. The rate is based on the current value of the underlying benchmark. This rate is used in the discount margin calculation to present value the cash flows.
ASW	The asset swap spread for the security, which is the difference between the coupon cash flow of the security and the fixed coupon of an offsetting swap in basis points. You can update curve settings for asset swap calculations in the *Asset Swap Calculator* (ASW) function. For more: ASW Help Page.

Term	Definition
Avg Life	The average life, which is the length of time (in years) it is expected to take to retire half of the debt through a sinking fund. For bonds with a sinking fund, if you select *Avg Life* as the yield to workout, the Street Convention, Equiv, and After Tax yields are calculated using the sinking fund provisions. They assume that: • When the bond is selling at a discount, the redemption calculation uses the market price. When the bond is selling at a premium, the redemption calculation uses par. • Only mandatory sinking fund payments are assumed in the calculations. • Any voluntary sinking fund payments that are exercised by the company are taken from the head or beginning of the schedule. The *Over Fund* field in the *Sinking Fund Analysis* (SF) function provides information regarding the number of bonds the issuer has retired above the mandatory amount. For more: SF Help Page.
B/M/A	For Australian floating rate notes, the swap rate. Defaults to Bloomberg generic composite rates from the *Interest Rate Swap Rates* (IRSB) function. The following is a list of ticker symbols of the indices used for the composite rate: Cash Rates – ACMR; Short Rates – BBSW1M, BBSW2M, BBSW3M, BBSW4M, BBSW5M, BBSW6M, BBSW9M, BBSW1Y; Swap Rates – ADSWAP1Q, ADSWAP2Q, ADSWAP3Q, ADSWAP4, ADSWAP5, ADSWAP7, and ADSWAP10. For more information, enter (ticker symbol) <INDEX> <GO>.
Basis	The interpolated CDS basis, which is the interpolated CDS rate minus the zero volatility spread.
Bench Bond	The type of price and yield precision in decimal places for the benchmark security.
Benchmark	The benchmark used to calculate the spread. • For floating rate notes, the security or index to which the security is linked, followed by the frequency of refix (if applicable).
Benchmark Bond, G-Sprd	A setting that allows you to choose the curve from which the benchmark bond is selected for the G-Spread calculation.
Benchmark Bond, I-Sprd	A setting that allows you to choose the swap curve from which the benchmark bond is selected for the I-Spread calculation.
Benchmark VTAM Weekly	For Venezuelan DPNs, the Tasa Activa de Mercados rate, which is Venezuela's benchmark rate and is released by the Central Bank of Venezuela every Thursday with a one-week delay. The default rate is from five days prior to the last coupon payment. For more information, enter VTAM <Index> <GO>.
Bid Px	The bid price of the security.
Bid Sz	The amount of the security the market maker is willing to buy at the bid price, where M = thousands and MM = millions.

Term	Definition
Blend	Allows you to calculate the blended average yield.
Blended	For Brady bonds, the conventional yield of cash flows to compare against non-Brady bonds.
Blended Yield	For Brady bonds, the conventional yield of the security's cash flows. No adjustments are made for any guarantees that may exist.
Blended Yield Analysis	• For callable bonds, allows you to calculate an aggregated yield for a bond that has multiple calls. Each call can be weighted by a percentage that you expect to be called on possible call dates. The resulting "blended" yield is calculated based on a risk-weighted blend of all yields to call. • For putable bonds, allows you to specify a put amount (expressed as a percentage) for a custom date, so that you can calculate an aggregated yield reflecting multiple puts. The resulting "blend" is calculated based on a risk-weighted blended average of the yields.
Bond	The type of price, yield precision (in decimal places), and price rounding for the loaded security.
Brady Bonds	A series of sovereign debt and complex credit-enhanced bonds that are issued under the umbrella of the Brady Plan Agreements. The name was coined from a speech by former U.S. Secretary of the Treasury, Nicholas Brady. In an effort to resolve the "LDC debt crisis", Mr. Brady offered U.S. Government multilateral support in obtaining debt and debt service relief for countries saddled with enormous foreign commercial bank debt. This help is available to nations who commit to a comprehensive program of fiscal reforms supported by the International Monetary Fund (IMF) and the World Bank.
Braess/Fangmeyer Yield	The yield calculated by discounting the payments received after a fraction of a year (from settlement to the next coupon) using simple interest, as opposed to compounded interest used in the conventional method. The Braess/Fangmeyer method is similar to the Treasury yield method for U.S. Treasury bonds and uses a 30/360 day type.
Breakeven P&L	For convertible bonds, the current profit or loss. It is the cash flow earned or lost if you convert the bond and sell the stock at its current market price. Breakeven P&L = [((conversion ratio * conversion price) – (bond price * 10)) – ((conversion price – current stock price) * conversion ratio)].
BTP ANN EQV	Appears for Italian floating rate notes. The bond equivalent of the Italian Treasury bond.
Buy	Allows you to launch an electronic inquiry buy ticket. For more: Ticketing a Trade.
Buy Price	For South Korean domestic bonds, the price per 10,000 Won.
Capital	The capital gains tax rate.
Capital Price	For Australian bonds, the clean price, without accrued interest.
Carry P and L	The cost of carry, calculated as (Interest Income – Finance Cost).

Term	Definition
Cash Require for Hedge	For convertible bonds, the net difference in the price of the bond less the price of the stock. For example: Cash = Bond Price – [(Stock Price * Conversion Ratio) * Hedge Ratio/100].
Cash Required/(XXX) Face	For convertible bonds, the additional amount of cash required to convert the bond. This occurs when the total price of the underlying shares is greater than the par value of the bond.
Cashflow	For convertible bonds, the period of time you can recoup the cash that was required to hedge at the annualized net cashflow rate.
Cashflow Last Update	For Danish mortgage bonds, the most recent date that cash flows are entered into Bloomberg Professional®. The cash flows are usually from the same day they are entered. **Note:** At the end of the day, Bloomberg stores the yield associated with the volume weighted average price as calculated by the Copenhagen Stock Exchange for each issue. (The history is stored in the <INDEX> database.) You can use the yield history in all graph and comparison functions to evaluate historical yields, yield spreads, dates, and more, back to December 1993. Danish Government bond yields are also stored this way. Bloomberg uses the seven digit ID code from the Copenhagen Stock Exchange as the ticker symbol. (The ID code is found on DES.)
Cashflows Revinvested @	Conventional yields are based on the internal rate of return (assuming all future coupons are reinvested at the internal rate), on an annual basis. If you enter a rate, the yield is recalculated (*Effective Yield*).
CD(ACT/[XX])	The day count basis used in the CD calculations and the CD yield equivalent.
Clean Price	The security's price minus the accrued interest.
Cnv Duration (yrs)	The present value weighted average maturity of the security's cash flows, in years. Also known as Macaulay Duration. For more: DUR Help Page.
Cnv, S/A	For Australian floating rate notes, the bond convention.
Commission	For South Korean domestic bonds, the percentage charged to execute the trade.
Comparable Mty	On the *Settings* screen, an option in the *Primary/Secondary Benchmark* drop-down menus. When selected, Bloomberg uses the bond from the currency matched benchmark curve with the maturity closest to the current issue as the default benchmark.
Compounded Coupon	For French variable coupon bonds, the current coupon compounded on the same basis as the coupon frequency. The coupon is calculated as follows: ((1+ Current Coupon/100^(1/number of payments per year)–1) rounded to 5 decimals. Then: The rounded value * 100 * number of payments per year.
Consensus	The Bloomberg Consensus benchmark. For more information, see the *Benchmark Rules* (BMRU) function and the BMRU Help Page.
Consortium Yield/ISMA Yield	The yield calculated using the day type and frequency of the bond and standard U.S./Euro market methodology with no adjustment for domestic market conventions. This yield is recommended by the ISMA.

Term	Definition
Conventional Annual Yield	For Danish mortgage bonds, yields always appear on an annual basis (convention in Denmark) regardless of the bonds coupon frequency.
Conversion Price	For convertible bonds, the price of the underlying stock used to calculate the number of shares received at conversion.
Conversion Ratio	For convertible bonds, the number of shares the investor receives upon converting one bond. **Note:** For convertibles with an underlying equity in a different currency, you must convert the par value of the bond into the underlying equity's currency. For example: (bond par value) / (bond currency exchange rate) = equivalent par value expressed in the equity currency. Then: (equivalent par value) / (conversion price) = conversion ratio
Convexity	The rate of change of the duration as yield changes. The second derivative of a security's price with respect to its yield, divided by the security's price. The security exhibits positive convexity when the price rises more for a downward move in the yield than the price declines for an equal upward move in the yield. Convexity measures the extent to which the price/yield curve for a bond is non-linear. Graphically, convexity measures the relative bend of a bond's price/yield curve while adjusted duration measures its relative slope. When large shifts in interest rates occur, bonds with positive convexity perform better than bonds with negative convexity.
Corporate Bond Equivalent	Indicates that the yield is calculated using semi-annual coupons with compounding in all periods and a 30/360 day type.
Cost	For warrants: the warrant price (expressed as a %) * par amount * number of warrants.
Cost of Carry (%)	The borrowing cost for the purchase of the bonds. The initial value is set to the corresponding bond currency's 6-month Treasury bill's bond equivalent rate.
Coupon Freq	The frequency of coupon payments per year.
Coupon Payments	• For multi-coupon bonds and toggle notes, the total value of coupon payments from the settlement date to the workout date. • For French variable coupon bonds, the current coupon multiplied by the number of bonds specified in the FACE AMOUNT field, in French francs.
Coupon/Day	For French variable coupon bonds, the daily coupon amount in French francs.
CPI @ Next Cpn Date	The projected consumer price index rate at the next coupon date.
CPI-Based Floaters	Consumer Price Index based floaters pay a variable coupon based on changes in the CPI-U index over the period plus one half of the quoted margin, adjusted for the number of days in the period.

Term	Definition
Cpn Accr (number of days)	For non-U.S. inflation-indexed bonds, the amount of interest, adjusted for inflation, accumulated, but not paid, between the issue date or most recent payment and the settlement date, calculated as Real Cpn Accrued Int * Index Ratio.
Cpn Frequency	For floating rate notes, the number of coupon payments per year. Coupon frequencies include annual, semi-annual, and quarterly.
Cpn Rate	For floating rate Brady bonds, the security or index to which the security is linked and the number of basis points to be added to it to derive the coupon rate.
Cpns Guaranteed	The coupons guaranteed. For Brady bonds, the coupons guaranteed are expressed as a percent of the face value of the bond. For fixed rate bonds, this percentage represents a fixed set of interest payments. For a floating rate note, this can represent more or fewer coupons, due to the floating nature of the bond.
Crystallized Rate	For French variable rate bonds, the last known reference rate, used for unknown cash flows. For more: French Variable Rate Bonds.
Currency	Appears for bonds that have been redenominated in Euros. You can choose to display the information in Euros or the legacy currency.
Current	The coupon/price, as a percentage.
Current (XX/XX) Cross Rate	For convertible bonds, appears when the bond and equity are in different currencies. The current exchange rate between the bond and equity.
Current Yield	The annual rate of return based on the price, calculated as [(stated dividend or coupon * par amount) /price] * 100. This is the actual income rate of return, as opposed to the coupon rate expressed as a percentage. For preferred securities, the annual dividend appears to the left of the yield on the *Calls* tab and in the *Yield to Call Analysis* (YTC) function.
Currently	For toggle notes, the coupon payment method in the current period, in cash or in kind.
Curve Convention	The yield to call and curve values according to the compounding frequency of the comparable government yield curve.
CUSIP	The Committee on Uniform Securities Identification Procedures number, which is a nine-character/digit code assigned by the CUSIP Service Bureau. It consists of a six-character prefix that identifies the issuer, followed by two characters/digits that identify the issue and a check digit.
Custom	In the *YTC* section, the yield to a date and price you enter.

Term	Definition
Date	• On the *Calls* tab, the call date on which a bond can be redeemed before maturity. • For multi–coupon bonds and floating rate notes, the rate reset date. For coupon rate history, access the *Security Description* (DES) function and select the *Floating Rates* menu option. For more: DES Help Page. • In the *Refix History* section for Australian floating rate notes, the coupon dates based on the settlement date, issue date, maturity date, and coupon frequency of the bond. • For U.K. index–linked stock, the date of the base and last reported retail price index values. • In the *Yield to Call Analysis* (YTC) function for preferred securities, the call date.
Days	The number of days to finance the trade/calculate the drop.
Days Accrued	For toggle notes, the amount of interest accumulated, but not paid, between the issue date or most recent payment and the settlement date. The number of days in the accrual period appears to the left of this field.
Days Accrued Int	For multi–coupon bonds, the interest accrued after a specific number of days.
Dealers Inventory	The quotes from dealer inventories.
DES - Description	A component on the *Custom* tab that shows issuer, coupon, and maturity information for the security using the *Description* (DES) function. The *DES - Description (medium)* option shows a larger panel with more descriptive information. For more: DES Help Page.
Dilution Protection	For convertible bonds, the amount of protection, expressed as a percentage, provided to the investor for equity corporate actions. For example, if there is a two for one stock split during the life of the convertible bond, a conversion price of $10 is adjusted to $5. This provides 100% protection to the investor against the diluted value of the underlying equity.
Discount Formula Equivalent	• For short term instruments, YAS uses the discount formula (T bill) method. • For Canadian T–bills, an ACT/365 day type method is used instead of ACT/360.

Term	Definition
Discount Margin	For floating rate notes, one of three ways of evaluating margins. This formulation has become the most widely accepted method. You can think of the discount margin (DM) as the margin relative to the base index rate, such that the present value of the cash flows equals the price plus accrued interest. Here, the discounting rate used is the assumed rate plus the discount margin. This formulation is very similar to the one used to evaluate a yield on a fixed income security. The major difference is that the first coupon period is discounted at the coupon rate fixed for that period (the index at next pay plus the discount margin). For a constant, unchanging index value, discount margin is the difference between the index and the monthly yield. Discount margin is quoted on the same basis as the instrument to which it is being applied.
Div. Pay Dt	The next dividend payment date.
Dlr	The pricing source code for the dealer.
DM	The discount margin. Appears when available.
Dollar Value of a (XXXX)	The change in price if the yield changes by [X] basis points. The greater the value of the basis point the greater dollar price volatility. **Note:** Price changes are almost symmetrical for small changes in yield, therefore it does not matter whether you increase or decrease yields to calculate this value. For larger changes in yield, however, there is a difference between the price value of a basis point if the yield is increased or decreased.
Domestic/International	For South Korean domestic bonds, the type of trade ticket pricing.
Duration	Also known as Macaulay Duration. The weighted average maturity of the security's cash flows, where the present values of the cash flows serve as the weights. The greater the duration of a security, the greater its percentage price volatility. The duration of zero coupon bonds is the maturity itself. For all other bonds, the duration is less than the maturity. • On the *Settings* screen, an option in the *Primary/Secondary Benchmark* menu. When selected, Bloomberg scans the list of bonds on the currency matched on-the-run benchmark curve and selects the security with the OAS duration closest to the security you are analyzing as the benchmark bond. • For convertible bonds, the present-value weighted average maturity of the convertible's cash flow, expressed in years. **Note:** This applies to the bond portion of the security and completely disregards the convertible option.

Term	Definition
Duration (Years)	The present value weighted average maturity of the convertible's cash flow, expressed in years. Also known as Macaulay Duration. **Note:** This only applies to the bond portion of the security and completely disregards the convertible option.
DV	The dollar value of a one basis point change in interest rates.
EDSF	Allows you to use a strip of Eurodollar futures contracts as a benchmark for spread analysis of short-dated bonds having a maturity that is shorter than 18 months. For more on Eurodollar strip contracts: EDSF Help Page.
Effective	The yield realized by investing coupon income at the specified reinvestment rate until maturity.
End	For multi-coupon bonds, the last date for the coupon rate.
Equiv	The street convention yield when changing the compounding frequency. This method uses the same day type conventions and the actual dates of the security's cash flows.
Equiv (number)/year	For multi-coupon bonds, a simplified yield conversion method (unlike government, corporate, and euro-bond equivalence). This method uses the same day type conventions and the actual dates of the security's cash flows.
Equiv (XX)/year Compound	• For Danish mortgage bonds, the conventional yield compounded (X) times per year. • For Australian bonds, a simplified yield conversion method (unlike government, corporate, and euro-bond equivalence). This method ignores both day count conventions and the actual dates of the bond's cash flows. • For non-U.S. inflation-indexed bonds, the yield, based on the selected frequency, compounding in all periods, and an actual/actual day count.
Equiv. Actuarial	For French variable coupon bonds, the bond equivalent rate of the crystallized rate, used to calculate the margin. For more: French Variable Rate Bonds.
Equivalent	For Brady bonds, the equivalent yield for a specified number of compounding periods per year.
Equivalent (compounding frequency)/Yr	The street convention yield when changing the compounding frequency. This method uses the same day type conventions and the actual dates of the security's cash flows.
Euro Bond Equivalent	The yield calculated using annual coupons with compounding in all periods and a 30/360 day type.
Euro Convention	Appears for Spanish domestic bonds only. Both price and accrued interest are calculated using an ACT/ACT day type.
Ex-Div Dt	The first date on which a security is traded without entitling the buyer to receive distributions previously declared.

Term	Definition
Ex-Dividend @ Settle	For South African governments, "Y" indicates that on or after the settlement date, the buyer of the bond is not entitled to the upcoming interest payment; therefore, the security trades with negative interest.
Exercise Price	For warrants, the price at which you can purchase the stock.
Face	The cash amount of the securities. Each (M) equals '000.
Face Amount	The cash amount of the securities. Each (M) equals '000.
Factor	The percentage of the original principal still outstanding. The original face amount multiplied by the factor equals the current amount outstanding.
First Cpn Dt	The first coupon payment date.
First Interest Payment On	For pay–in–kind bonds, the date the first interest payment is due.
Flat	For non–U.S. inflation–indexed bonds, Face * Flat Index Ratio * Price.
FPA - Financing	A component on the *Custom* tab that displays repurchase and cost of carry data for the selected security.
French Number	For French variable coupon bonds, the French identification number of the bond.
Freq	For Australian floating rate notes, the coupon frequency of the swap.
Fwd Prc	The fair price of the security for future settlement, given the repo rate and the coupon income, calculated as (Market Price – (Interest Income – Finance Cost)).
Gearing	For warrants, the leverage of the warrant with respect to the stock price. Leverage is calculated as: stock price / (warrant price/# shares)
Gilt True Yield Equivalent	For U.K. strips, calculated using standard compounding and an act/365 day count.
GIP - Intraday Price Chart	A component on the *Custom* tab that allows you to track the intraday price of the security over the last two weeks. The *GIP - Intraday Price Chart (medium)* option shows a larger panel with a larger chart.
Gives an Effective Yield of	• For multi–coupon bonds, the return when all the cash flows are reinvested at the stated reinvestment rate (@workout). The interest on interest is the interest that is earned by investing the coupon payments at the reinvestment rate. • For Danish mortgage bonds, the yield derived by incorporating the reinvestment rate, on an annual basis.
GP - Historical Line Graph	A component on the *Custom* tab that shows a graph of price, yield, or spread for the selected security. The *GP - Historical Line Graph (large)* option shows a larger panel with a larger chart. The drop–down menus at the top of both components allow you to select the metric and time frame that appear in the chart.
Gross Amount	For non–U.S. inflation–indexed bonds, Flat + Inflation Accrual.
Gross Dom Yield	For floating rate notes, the yield on an annual compound basis, without taxes, and with no adjustments for coupon dates falling on weekends or holidays.

Term	Definition
Gross Price	For Australian bonds, the full price, including accrued interest.
Gross Profit	For multi-coupon bonds, toggle notes, and Australian bonds, the difference between total income and total payment.
Gross Semi-Annual True Yield	For Italian Governments, the yield to maturity, before taxes, compounded semi-annually.
Gross True Yield	For Italian Governments, the yield before taxes, adjusted for weekends and holidays.
Gross Yield	For Italian Governments, the yield to maturity, before taxes.
G-Sprd	The G-spread, which is the interpolated bond spread to the corresponding benchmark curve. The G-Spread is calculated by subtracting the interpolated government bond yield from the selected bond's yield to workout.
G-Spread	The interpolated bond spread to the corresponding benchmark curve. The G-Spread is calculated by subtracting the interpolated government bond yield from the selected bond's yield to workout.
Habitual	For South Korean domestic bonds, the same as Treasury Yield. The number is linear in the first period, compounds in others, and uses an actual/actual day type.
Haircut (%)	The percentage difference between the market value of the stock position that is collateralized and the dollar proceeds lent out or borrowed. The shorted stock position is covered by borrowing stock and paying out the market value, in addition to some percentage of the proceeds.
Hedge Ratio (%)	For convertible bonds, the number of bonds bought or sold against a position in the underlying security to hedge the position, expressed as a percentage.
Holding Period	For South Korean domestic bonds, the purchase date (included) to the selling date (or redemption date, disregarded). Leap month is included in case the holding period is less than one year.
HS - Spread Analysis	A component available on the *Custom* tab that plots the historical spread between a security and its benchmark over the time period you select. The top chart tracks the yields of each security, while the bottom chart tracks the spread between them.
ID - Identifiers	A component on the *Custom* tab that shows various identifiers for the selected security, such as the CUSIP and ISIN numbers. These identifiers are drawn from the *Description* (DES) function. For more: DES Help Page.
Inc/Ret - Income Return	A component on the *Custom* tab that allows you to track both the income made on a security investment and the return generated from that investment, based on the yield calculated by YAS and face amount you specify.
Income (XX)%	The income tax rate.
Index Assumption	Applicable to inflation-linked bonds that are linked to Brazil IPCA or IGP-M. Also known as the IPCA assumption or IGP-M assumption. The *Index Assumption* is included in the calculation of the index variation for current settlement. For more on IPCA methodology and calc types 1193 and 1195: CALT Help Page > (1193) Brazil ILB - Brazil Inflation Linked Securities, CALT Help Page > (1195) Brazil ILB - Brazil Inflation Linked Securities.

Term	Definition
Index to (date)	For floating rate notes, the next coupon date and assumed benchmark rate for the time period from settlement to next coupon. This rate is used only to present value the current coupon in the discount margin calculation. If the number of days between settlement date and the next payment date is less than the refix frequency of the floater, the assumed index may be inappropriate for present valuing the next coupon payment. **Note:** This field allows you to employ a more appropriate rate for the next coupon payment in the discount margin calculation.
Index Variation	Applicable to inflation-linked bonds linked to Brazil IPCA or IGP-M. The inflation adjustment value for current settlement. For more on IPCA methodology and calc types 1193 and 1195: CALT Help Page > (1193) Brazil ILB - Brazil Inflation Linked Securities, CALT Help Page > (1195) Brazil ILB - Brazil Inflation Linked Securities.
Inflation Accrual	For non-U.S. inflation-indexed bonds, the Accrued Ratio Growth multiplied by the Price and the Face.
Inflation Adj Duration	For U.K. index-linked stock, the percentage price change of a security for a given change in the money yield. The higher the modified duration of a security, the higher its risk. Adj/Mod Duration = [duration/{1 + (IRR/M)}]; where IRR is the internal rate of return and M is the number of compounding periods per year.
Inflation Assumption	Applicable to inflation-linked bonds. Defaults to a rolling 12-month annualized change in the reference CPI value. The reference CPI for a particular bond can be an interpolated value or the latest known CPI value for a given settlement date, depending on the structure of the security. For UK eight month lag bonds (calc type 44), the *Inflation Assumption* is 3%, per DMO methodology. For Israel bonds (calc type 1238), the *Inflation Assumption* is the corresponding Bank of Israel CPI forecast: ISIFAR Index. For more on calc types and their reference CPIs: CALT Help Page.
Inflation Compensation	For non-U.S. inflation-indexed bonds, the increase in the principal value of the securities based on the change in the reference CPI inflation since issue, calculated as Face * (Index Ratio – 1).
Interest @ (reinvestment rate)	The interest calculated by investing the coupon income at the reinvestment rate (reinvestment income).
Interest Margin	For Australian floating rate notes, the amount added to or subtracted from the Bank Bill Swap (BBSW) reference rate to set interest payments, in basis points.
Interest on Interest	For multi-coupon bonds, the interest calculated by investing the coupon income at the reinvestment rate (reinvestment income).
Interest Rate	For Brady bonds, the interest rate sensitivity, e.g., the impact of changes in the underlying U.S. Treasury curves.
Interest Yield	For U.K. index-linked stock, the coupon rate divided by the clean price.
Interpolated Yield	The yield of maturities that fall between the specific maturities represented by securities in a yield curve. The interpolated yield is derived using methodologies such as bootstrapping and regressions.
Interpolated/Specific	In the *Refix History* section for Australian floating rate notes, the swap type.

Term	Definition
Invoice - Buy/Sell Ticket	A component on the *Custom* tab that calculates the total amount of money required to settle a trade for the specified face amount of the security and the selected settlement date.
I-Sprd	The I–Spread, which is the interpolated bond spread to the benchmark swap curve. The I–Spread is calculated by subtracting the interpolated swap rate from the selected bond's yield to workout.
I-Spread	The interpolated bond spread to the benchmark swap curve. The I–Spread is calculated by subtracting the interpolated swap rate from the selected bond's yield to workout.
Issue Date	For toggle notes, the first settlement date of the bond.
Japanese Yield (Simple/Compound)	The Japanese convention of evaluating bonds on a simple or compound interest basis. You can select simple or compound interest from the *Bond Calculation Override* (COVR) function. For more: COVR Help Page. • Using the simple interest basis, the effects of compound interest are ignored. Simple yield is the annualized cashflow as a percent of the original flat price. The Japanese equivalent simple yield appears in the same position as the proceeds yield, when it is not calculated (e.g., more than two years to maturity). • Japanese compound yield includes the future value of the current price plus accrued interest of a bond invested to maturity, plus the total face value of the bond and the coupon incomes, plus the reinvestment income of the coupons.
Last Refix Date	For Australian floating rate notes, the coupon's last reset date.
Leap Year	For South Korean domestic bonds, leap year is disregarded (a year is always 365 days).
M/M Equiv To Next Fix	The money market equivalent values the debt as a short–term security maturing at the next refix date and calculates a yield.
Margin	For French variable coupon bonds, the spread over the reference index. This is the yield less the actuarial crystallized rate. For more: French Variable Rate Bonds.

Term	Definition
Maturity	• When selected from the drop-down menu next to the *Yield* field, Bloomberg uses the price from the default settings for bonds with redemption provisions to compute a yield-to-worst workout date, then uses a bond from the currency matched on-the-run benchmark issues that matches the current workout date. • In the *YTC* section, the maturity date, price, and yield to maturity. Yield to maturity is the percentage rate of return paid if the security is held to its maturity date. The calculation is based on the coupon rate, length of time to maturity, and market price. It assumes that coupon interest paid over the life of the security is reinvested at the same rate. • For multi-coupon bonds/floating rate notes, the date that the principal amount is due and payable.
Medium Term CD (Act/360)	The money market equivalent yield that equates a periodic coupon-paying security to a security that pays interest at maturity. All coupon payments received use simple interest compounded to the next pay date, and so on until maturity. Then, the total value at maturity is present valued to the settlement date on a simple-interest basis. Only valid for up to two years from maturity. Canadian governments and U.K. Gilts use an ACT/365 day type.
Message Quotes	The quotes from messages. You can specify whether you want the *Quotes Manager* section to display your personal, firm, or shared messages only.
Mmkt (Act/)	The money market equivalent yield that equates a periodic coupon-paying security to a security that pays interest at maturity. Each cash flow is compounded from its coupon payment date to the next payment date, and so on until the last payment date. The total is present valued back to the present date via simple interest. An ACT/360 day type is used. The drop-down menu to the right of the *MMYld* field allows you to specify the yield that appears in the field to its right. • For Canadian Government Bonds and U.K. Gilts, the yield is calculated using an ACT/365 day type. This value may not be valid when the remaining time to maturity is more than two years. • For Australian floating rate notes, the day count and rate.
Modified Duration	The percentage price change of a security for a given change in yield. The higher the modified duration of a security, the higher its risk. Modified duration is calculated as [duration / {1 + (IRR/M)}]; where IRR is the internal rate of return and M is the number of compounding periods per year.
Money Yield	For U.K. index-linked stock, the yield obtained after discounting the inflation assumption from the inflation adjusted cash flows. This yield incorporates the three month lag impact and prior inflation adjustment made on the bond.

Term	Definition
Moosmueller Yield	The yield calculated by discounting the payments received after a fraction of a year (from settlement to the next coupon) using simple interest, as opposed to compounded interest used in the conventional method. Moosmueller yield uses money market discounting from the next coupon date back to the settlement date.
Net Amount	• For CPI–based bonds, the total amount of the invoice payment. Calculated as principal + accrued interest. • For non–U.S. inflation–indexed bonds, the net value of the payment invoice, including inflation adjustments that have been made to the accrued interest and face amounts.
Net Cash Flow ($)/Yr	For convertible bonds, the annualized cash flow, in dollars per year per bond, for buying the bond and financing it at the cost of carry rate, as well as selling the stock short and reinvesting the proceeds at the short rebate rate. The shorted stock is covered by borrowing stock and paying the market value plus some haircut. It is necessary to finance the haircut dollar value at the cost of carry rate.
Net Semi-Annual True Yield	For Italian Governments, the yield to maturity, after taxes, compounded semi–annually.
Net True Yield	For Italian Governments, the yield after taxes, adjusted for weekends and holidays.
Net Yield	• For Italian Governments, the yield to maturity, after taxes. • For South Korean domestic bonds, the after–tax yield for coupon income derived by deducting the tax from the habitual yield.
Neutral Price	For floating rate notes, the price that generates the same discount margin, but for settlement at the next coupon date rather than the settlement date at the top of the screen. This calculation allows an investor to sell a floater for settlement after the next coupon is collected. The neutral price represents the price of the security for a roll on the next coupon date.

Term	Definition
Next Cpn Date	• For Australian floating rate notes, the date of the next coupon payment. • For Venezuelan DPNs, the next coupon payment date. This date is determined based on the maturity. For example, if the maturity date is 12/13/04, the coupon is paid on the 13th of each quarter. **Note:** The number of actual days must be calculated using ACT/360 day type if the bond was issued after February 20, 2001. • For toggle notes, the next interest payment date.
Next Ex-Draw Date	For Danish mortgage bonds, the date the bond trades ex–drawing. If you purchase a bond in the ex–drawing period, you are not entitled to the scheduled repayment of principal on the forthcoming coupon date. **Note:** If the next ex-draw date is before the next coupon date, the bond is not trading ex-drawing; therefore, you can participate in the next drawing. In this case, the ex-draw date is the actual date announced by the Copenhagen Stock Exchange. If the next ex-draw date is after the next coupon date, the bond is trading ex-drawing; therefore, you cannot participate in the next scheduled repayment. In this case, the ex-draw date is the next assumed draw date. Once this assumed draw date appears, all future ex-draw dates are assumed and not actual. Bloomberg automatically updates to the actual ex–draw date, which is usually one working day after the coupon date.
Next Interest Payment	For U.K. index–linked stock, the cash flow to be paid on the next coupon date, expressed as a percentage of the face amount.
Next Negative AI Date (date) (days)	For Australian bonds, the next negative accrued interest date. If you buy the bond on or after the (date), you do not receive the interest that is paid in (days), hence you purchase the security with negative interest.
Next Pay	For floating rate notes, the date on which the next coupon is paid.
Next Refix Date	For Australian floating rate notes, the coupon's next reset date.
Number of Bonds	For multi–coupon bonds, the number of securities, where each security equals a face value of 1,000.
OAS	The option–adjusted spread, which is the incremental return of the security compared to a benchmark interest rate curve, adjusted for embedded options.
Overnight Repo Equivalent	The yield calculated with interest accumulated over one night, holding price fixed, and using a CD ACT/360 day type.
P&L	The profit or loss generated by the immediate exercise of the warrant. P&L = Percent Premium * (Stock Price/100)

Term	Definition
Parity	For convertible bonds, the bond price at which neither a profit nor a loss is realized (ignoring transaction costs). This assumes you convert the bond upon settlement and sell the stock immediately at its market price. For example: Parity price = (current stock price * conversion ratio) / 10. **Note:** A bond price lower than the parity price implies you can buy the convertible bond and convert it for an immediate profit. Therefore, the parity price acts as the floor for the bond price.
Pay-In-Kind Redemption Value Growth Factor	For pay–in–kind bonds, the factor by which the principal grows between the settlement date and the first interest payment date.
Percent Premium	• For convertible bonds, the premium points expressed as a percentage. For example: Percent premium = (premium points/bond parity price) * 100 **Note:** A low percent premium implies you pay little for the fixed income protection and the bond is sensitive to fluctuations in the stock price. A high premium implies you pay more for the fixed income protection, which makes the bond more sensitive to interest rate changes. • For warrants, the warrant's cost over the stock price. The formula: (((warrant price/# shares + exercise price)/stock price) –1) * 100
PIK Premium	For toggle notes, the spread between the pay–in–kind coupon rate and the cash coupon rate.
PIK Yield to {date}	For pay–in–kind bonds, the yield calculated to a specified date, other than the scheduled first real coupon payment date.
PIK Yield to Maturity	For pay–in–kind bonds, the yield that assumes the holder keeps the additional bonds they receive, resulting in the holding increasing by the Redemption Value Growth Factor.
Points	For convertible bonds, the dollar amount, in price points, that you pay above the value of the equity for the fixed income protection afforded by the convertible bond. For example: Premium points = (current bond price – bond parity price).
Prepay	For Danish mortgage bonds, the percentage of your holding that is pre–paid on each coupon date.
Prev Cpn Date	• For floating rate notes and Venezuelan DPNs, the date on which the current coupon period begins. • For toggle notes, the previous interest payment date.

Term	Definition
Price	The ask price of the security, expressed as a percentage of par. The price rounding amount for the corresponding benchmark appears in the highlighted field to the right of the price. The timestamp for the calculation appears to the right of the price rounding amount. If you change the price, all yields are recalculated along with duration and yield value of a 32nd. • For convertible bonds, the price of the underlying stock. • For French variable coupon bonds, the clean price of the bond. The clean price does not include any accrued interest that the seller is entitled to receive. • For bonds in default, the purchase price of the bond. • For toggle notes, the default price of the bond. The assumptions underlying the default price/yield calculations are: the primary price defaults to the all cash price when the last coupon payment was cash, or all pay–in–kind (PIK) when the last payment was made in kind. If the bond is in a PIK period, but is scheduled to pay cash starting from a specific date, the default price reflects the combination of PIK and cash pay. • On the *Yield to Call Analysis* (YTC) function for preferred securities, in the calculator inputs section, the current ask price of the security, according to the pricing source you selected for the security. You can select a pricing source on the *Price Source Selection* (PCS) function. For more: PCS Help Page. • On YTC for preferred securities, in the yield table section, the call price of the security on the specified date.
Price @ Fix	The price of the security on the refix date. **Note:** This price is only used for the money market equivalent calculations. It has no impact on the margin calculations.
Price at Refix	The price on the next refix date.
Price Drop	The price drop, calculated as (Settlement Price – Forward Price).

Term	Definition
Price Value of a (yield change)	The change in price of a security if yield changes by a given number of basis points. The greater the basis point change, the greater the dollar price volatility of the security. **Note:** Price changes are almost symmetrical for small changes in yield; therefore, it does not matter whether you increase or decrease yields to calculate this value. For larger changes in yield, there is a difference between the price value if the yield is increased or decreased. The change in price value depends on the convexity of the security.
Principal	The principal amount, calculated as the price multiplied by the face amount.
Proceeds Hedge	The hedge ratio multiplied by the face amount of the corresponding security. The hedge ratio is calculated as [total money(security)/total money(hedge bond)].
Proceeds Mmkt Equivalent Net	For Italian Governments, the money market equivalent yield, after taxes, that is determined only when the remaining time to maturity is less than two years.
Proceeds/Mmkt	The Proceeds/Mmkt Equivalent, which is the money market equivalent yield that equates a periodic coupon–paying security to a security that pays interest at maturity. Each cash flow is compounded from its coupon payment date to the next payment date and so on until the last payment date. The total is present valued back to the present date via simple interest. An ACT/360 day type is used. **Note:** For Canadian Government Bonds and U.K. Gilts, the yield is calculated using an ACT/365 day type. This value may not be valid when the remaining time to maturity is more than two years.
Proceeds/MMkt (Act/365)	For South African governments, the money market equivalent yield that is determined only when the remaining time to maturity is less than two years.
Proceeds/Mmkt Equivalent Gross	For Italian Governments, the money market equivalent yield, before taxes, that is determined only when the remaining time to maturity is less than two years.
Projected Coupon	For CPI–based floaters, the coupon is projected for the next payment date based upon the known CPI values and the inflation assumption.
Projected CPI Component	For CPI–based floaters, 100 * (CPI next cpn – CPI last cpn) / CPI last cpn.
Projected Dividend	Appears for floaters. The projected next dividend.
Provisional Hedge	For convertible bonds, the ratio of the dollar advantage obtained from the conversion at the provisional price to the difference between the provisional price and the current equity price, times the conversion ratio.
Provisional Price	For convertible bonds, the market price that the underlying security must reach, for a specified period of time, to prompt a special call redemption. The special call is usually at a different redemption price that the normal redemption schedule.
PV of Interest Guarantee	For Brady bonds, the present value of interest guaranteed.
PV of Principal Guarantee	For Brady bonds, the present value of principal guaranteed, the percent of principal that is guaranteed (secured), and the discount rate (defaults to the appropriate strip rate). If the bond has guaranteed rolling interest payments, they are applied to the next nearest payment after settlement date.

Term	Definition
QMGR - Quotes Manager	A component on the *Custom* tab that allows you to monitor all the prices you are receiving for the selected security via messages and/or dealer inventory pages. The *QMGR -Quotes Manager (wide)* option provides a wider panel with more quote information, while the *QMGR -Quotes Manager (long)* option lists a longer set of quotes. For more: QMGR Help Page.
QR - Quote Recap	A component on the *Custom* tab that displays an intraday quote recap for the selected security from TRACE or the current price source. The *QR - Quote Recap (medium)* option provides a longer panel with more price ticks. Both components show data from the *Quote Recap* (QR) function. For more: QR Help Page.
Quoted In	For convertible bonds, appears when the bond and equity are in different currencies. The currency in which the bond is quoted, if applicable.
Quoted Margin	• For floating rate notes, the spread, in basis points, added to or subtracted from the benchmark rate. • For French variable coupon bonds, the margin used to fix coupons relative to the reference index, in basis points.
Quoted Margin Over CPI-U	For CPI–based floaters, the widely accepted method for calculating the discount margin (=DM) of a floater is to generate a projected cash flow using an assumed average value (=AR) for the index over the remaining life of the bond and discounting them back with a rate equal to the AR + DM, such that the present value of the cash flow equals the price plus the accrued interest.
R	The reporting party side.
Rate	• For multi–coupon bonds and floating rate notes, the coupon rate for the specified period. Many floating rate notes are subject to caps and/or floors when coupons are reset. YAS takes these restrictions into account. • In the *Refix History* section for Australian floating rate notes, the spot rate.
Rate to Next Fix	For Australian floating rate notes, the assumed financing rate from settlement to the next refix date. This rate is used only as the financing rate in the adjusted price calculation.
Rating - Bond Ratings	A component on the *Custom* tab that lists the current ratings for the bond from the major credit reporting agencies.

Term	Definition
Real (Effective) Spread	For CPI–based floaters, a similar concept has been developed at Bloomberg to analyze the resulting real component of the spread earned over the life of the security, using an assumed inflation rate. For these securities, the coupons are set to be equal to some fixed spread (=FS) plus the change in CPI over the coupon period (=DI), obtained from the assumed inflation rate. A projected cash flow is discounted back with a rate equal to the average value (=ADI) of the CPI component of all coupons (DI), plus some average fixed spread earned (=AFS), and is equal to the present value. The yield of this cash flow is the money yield (MY). Therefore, Bloomberg sets MY = AFS + ADI and solves for AFS, the average fixed spread earned over the life of the bond, which Bloomberg calls the Real (Effective) Spread.
Real Coupon	For non–U.S. inflation–index bonds, the stated interest rate of the security, without inflation assumptions.
Real Cpn Accrued Int	For non–U.S. inflation–index bonds, the accrued interest since the last coupon payment date.
Real Yield	For U.K. index–linked stock, the nominal return. The return an investor would require on a normal bond that would make the real return equal to that of the inflation–indexed bond, assuming a constant inflation rate.
Real Yield Adj Duration	For U.K. index–linked stock, the percentage price change of the security for a given change in the real yield. The higher the modified duration of a security, the higher its risk. Calculated as: Adj/Mod Duration = [duration / {1 + (IRR/M)}], where IRR is the internal rate of return and M is the number of compounding periods per year.
Redemption	• For floating rate notes, the price paid for the principal amount on the redemption date. • For Venezuelan DPNs, the price paid for the principal amount on the redemption date. This is usually 100 percent of face value.
Redemption Value	For multi–coupon bonds and toggle notes, the value of the security at the workout date.
Reference Percent	For Venezuelan DPNs, the reference percentage (Porcentaje de Referencia) used to calculate the floating coupon rates. The reference percentage is set by the Central Bank of Venezuela on the security's placement date and does not change.
Reference Period	For French variable coupon bonds, the dates for the time period used to calculate the coupon.
Reference Rate	For French variable coupon bonds, the reference index and rate used to calculate the coupon.
Refix Freq	For floating rate notes, the number of times the rate resets per year.
Repo Rate	The rate at which you can borrow money to purchase the corresponding security. The three–month money market rate appears in the *Repo Rate* field by default. You can find global money market rates in the *Treasury & Money Markets* (BTMM) function. For more: BTMM Help Page.

Term	Definition
Repo to (date)	For floating rate notes, the next refix date and assumed financing rate for the time period from settlement to next refix. This rate is used only as the financing rate in the adjusted price calculation.
Return	The sum of interest, principal payments, reinvestment income, and any capital gains/loss annualized, expressed as a percentage. For Italian Governments, the total return is based on semi-annual compounding.
Risk	A measurement used by Bloomberg to indicate price sensitivity given shifts in interest rates. Risk is 100 times the price value of a basis point change in yield. For Italian Governments, the.01 is based on the yield chosen (conventional or semi-annual equivalent). **Note:** Risk times 100 divided by the full price is equal to adjusted duration. The full price includes accrued interest. Risk is calculated to a particular workout date. The value in the left column displays maturity. • Fix to Floater takes into account current rate information for benchmarks that are used during the floater period, as well as the current fixed rate information. • On the *Yield to Call* tab, the risk is calculated assuming redemption on each call date, using semi-equivalent yields as the benchmark.
Risk - Risk and Hedge	A component on the *Custom* tab that shows modified duration, risk, and convexity data with the traditional yield and OAS methodology. The selected benchmark is used as the hedge security. If the selected security is option-free, the OAS risk measures are similar to the maturity measures. If the selected security is callable and in-the-money, traditional workout and OAS risk measures can produce different hedge measures.
Risk Factor	• For bonds in default, the probability that an investment will earn the amount of profit that it is supposed to earn, or the chance an investment will drop in value. Risk is also an indication of price sensitivity given shifts in interest rates. Bloomberg risk is 100 times the value of.01, where the.01 is based on the type of yield. • On the *Yield to Call Analysis* (YTC) function for preferred securities, a measurement used by Bloomberg to indicate price sensitivity given shifts in interest rates. Risk is 100 times the price value of a basis point change in yield. It is calculated assuming redemption on each call date, using semi-equivalent yields as the benchmark.
Risk Hedge	The risk hedge or workout hedge is the risk of the bond divided by the risk of the benchmark, multiplied by the face amount.

Term	Definition
Risk/Inv - Risk and Invoice	A component on the *Custom* tab that displays several measures of risk, so you can hedge the security under analysis and gauge its appropriateness for your investment strategy. For the specified face amount of bonds and the selected settlement date, the *Risk and Invoice* component calculates the total amount of money required to settle a trade.
RPI	For U.K. index–linked stock, the current and base retail price index value at the security's issuance. The general Retail Price Index (RPI) measures the average change from month to month in the prices of goods and services purchased by households in the United Kingdom. RPI is the main domestic measure of inflation in the U.K. The spending pattern on which it is based is revised each year using information from the Family Expenditure Survey. For more information, enter UKRPI <Index> DES <GO>.
Sched	For Danish mortgage bonds, the scheduled percentage of your holding that is paid on each coupon date.
Sched & Unsched	For Danish mortgage bonds, payments based on the same cashflows as scheduled payments but adjusted for the Prepayment Speed Assumptions you select. If you use the CPR and/or SPR prepayment models, YA recalculates the sensitivity measures.
Scheduled Payments	For Danish mortgage bonds, payments solely based on the cash flows as published by the Copenhagen Stock Exchange and do not take into account any prepayments. Cash flows are downloaded daily into the Bloomberg Professional® database (but not stored historically).
Secured at	For Brady bonds, an interpolated rate off of the curve for the maturity date. This is the rate at which the principal is present valued for stripping, in percent.
Security Convention	The yield to call and curve values according to the compounding frequency of the security.
Sell	Allows you to launch an electronic inquiry sell ticket. For more: Ticketing a Trade.
Settle	The date(s) on which the corresponding securities must be delivered and paid to complete a transaction. On the *Custom* tab, you can add the FPA - Financing component, in which the *Settle* field displays the amount required to settle the initial leg of the financing trade.
Settlement	The date securities must be delivered and paid for to complete a transaction.
Settlement Date	The date the securities must be delivered and paid for to complete the transaction.
Shares/Warrant	The number of shares per warrant.
Short Rebate (%)	The assumed reinvestment rate of proceeds resulting from the short sale of stocks. This rate is used in the *Break-Even* and *Net Cash Flow* fields. The initial default value is the Broker Loan rate.
Simple Interest (Act/360)	Appears for discount securities only. The annualized cash flow expressed as a percentage of the original flat price.
Simple/Compound Yield to Workout	For convertible bonds, appears for Japanese domestic convertibles only. The simple or compound yield to the workout date.
Sinks/Principal	For Brady bonds, the percent of the sinks/principal that is guaranteed.

Term	Definition
Size	• In the *Size* drop–down menu, the size requirement for price quotes. • In the *Quotes* table, the size of the trade.
Sovereign Spread	For Brady bonds, the value of a constant spread over the U.S. Treasury rate that equates the present value of the modified cashflow to the net investment value that is subject to sovereign risk, in basis points.
Spd. Risk	The spread risk, which is the change in price based on a 1 basis point change in Z-DM.
SPRD - Spread to Curve	A component on the *Custom* tab that displays the spread between the workout yield of the selected security and the interpolated yield of a benchmark curve in the corresponding currency. You can select the benchmark curve from the drop–down menu at the top of the component.
Sprd/Yld - Spread/Yield	A component on the *Custom* tab that displays the difference between the yield to the workout date and the yield interpolated to the matching point on various benchmark curves in the corresponding currency, so you can assess the relative value of the security under analysis. It also displays bond yields using multiple compounding conventions, allowing you to assess the relative value of the security under analysis. The general format used by YAS displays the standard yield value for the security as the first yield value, followed by several equivalent yields.
Spread	The basis point difference in yield between the security under analysis and the hedge bond (or benchmark bond). For Brady bonds, indicates the spread sensitivity, e.g., the impact of changes in the sovereign spread, given spread duration, risk, etc.
Spread @ Curve	For Brady bonds, the spread of the stripped yield (in basis points) to the yield of the guarantee curve, interpolated to the maturity date.
Spread Component	For CPI–based floaters, calculated as: quoted margin * (# days next cpn – previous cpn) / actual number of days per year.
Spread for Life	A method commonly used to evaluate Sallie Mae FRNs. The spread for life consists of two components: the spread to the T–bill rate at which the coupon is fixed (quoted margin) and the capital gain or loss realized at maturity. This gain or loss is the difference between the purchase price and par, and is converted to a percentage change, then discounted for the remaining life of the FRN. Spread for life does not include accrued interest. To include accrued interest, enter the price plus accrued interest in the *Price* field instead of the clean price.
Spread Over Treasury Curve	For Brady bonds, the spread over the U.S. Treasury curve, in basis points. The spread measures the difference between the blended yield and the U.S. Treasury rate respective to the bond's average life.
Src	The price source.
Start	For multi–coupon bonds, the initial date for the coupon rate.
Stock	For convertible bonds, the ticker symbol of the underlying security appears.

Term	Definition
Stock Price	For warrants, the current stock price, quoted in the currency in which the stock is quoted. **Note:** The stock and exercise prices are denominated in the stock's currency.
Street Convention	The yield calculated according to the conventions used in the market the bond was issued in. For Brady bonds, includes all of the cash flows contained in the bond.
Street Real Yield	For non-U.S. inflation-indexed bonds, the standard yield-to-maturity calculation.
Strip Price	For Brady bonds, the price of the bond stripped of the present value of its guaranteed components.
Strip True Yield Equivalent	For U.K. strips, indicates that the yield is calculated with coupon dates moved from a weekend or holiday to the next valid settlement date.
Strip Yield	The annual rate of return based on the strip price. It is calculated as [(stated dividend or coupon * par amount) /strip price] * 100. The strip price, which appears to the left of the strip yield, is the current price minus accrued interest.
Stripped	Indicates the bond is stripped of its guaranteed parts (yield). The stripped yield is the yield of the uncollateralized cash flow to the strip price.
Stripped Yield	For Brady bonds, the sovereign discount rate that equates the present value of the modified cash flow generated by the model to the net investment. Stripped yield represents the return on pure sovereign risk for both collateralized and uncollateralized bonds.
SWAP	In the *Refix History* section for Australian floating rate notes, the frequency and swap rate.
Tax: Base Price	The base price, which is equal to the issue price of the bond.
TED	The Euro-Futures strip spread, which is the spread between the security and a Euro-Futures strip with matching maturity.
Term	The termination money due at maturity of the repo, calculated as (Settlement Amount + Finance Cost). • For Australian floating rate notes, the term length for the swap.
Theoretical	For South Korean domestic bonds, the same as Street Convention. The standard yield to maturity calculation for the security.
Time	The time the price was quoted.
To	The workout date, which is the most likely redemption date for the security given the current price and redemption schedule.
Toggle notes	The bonds that give the issuer the option to pay the coupons in cash or in additional bonds. The latter methodology is also known as PIK (Pay In Kind). When a bond "PIKs," additional bonds are issued in lieu of cash payments. Also, bonds that are in the PIK period trade without accrued interest.

Term	Definition
Tot/Ret - Total Return	A component on the *Custom* tab that allows you to calculate the total return from investing in the security over the past month, quarter, and year, as well as over a custom horizon period. You can specify the pricing source and market side on which to base the calculation.
Total	The total value of all income, calculated as redemption value plus coupon payment plus reinvestment income. In the *Invoice* section, indicates the total value of the payment invoice, calculated as principal plus accrued interest. The currency appears to the right of the field.
Total Income	For multi–coupon bonds, the total value of all income, calculated as redemption value plus coupon payment plus reinvestment income.
Total Payment	For multi–coupon bonds, the total value of the payment invoice, calculated as principal plus accrued interest.
Traded Margin (BPS)	For Australian floating rate notes, the margin at which the security is trading over the Bank Bill Swap (BBSW) reference rate, in basis points.
Traded Margin Risk	For Australian floating rate notes, the risk, expressed in basis points, associated with a shift in the traded margin.
Treas Crv	The comparable government yield.
Treas Curve	The comparable treasury yield.
Treas Sprd	The spread between the Yield and Treas Crv.
Treas Spread	The spread between the Yield and Treas Curve.
Treasury Yield	Used by the Treasury Department on new issues as a comparison with when–issueds. The number is linear in the first period, compounding in others, using an ACT/ACT day type.
True Annual Yield	For Danish mortgage bonds, calculated on the same basis as conventional yield, except that cashflows are bumped to the following work day if the coupon date falls on a weekend.
True Net Yield	For Italian floating rate notes, the yield on an annual compound basis with adjustments both for taxes and for coupon dates that fall on weekends or holidays.
True Yield	The yield calculated with coupon dates moved from a weekend or holiday to the next valid settlement date.
U.S. Govt Equivalent	For multi–coupon bonds, South African governments, and U.K. strips, the yield calculated using semi–annual coupons in all periods and an ACT/ACT day type.
UK Strip Convention	Uses standard yield to maturity calculations with compounding for all periods.
UK Strips	Securities stripped from U.K. Treasury Gilts, but the calculations are different from U.K. Gilts. Strips are calculated with an ACT/ACT calendar, no ex–dividend, and always compound in the yield calculation. U.K. Gilts use an ACT/365 calendar, go ex–dividend, and do a True yield calculation, which switches to simple interest in the last coupon period before maturity.
Using a Reinvestment Rate of	For multi–coupon bonds, the rate at which coupon payments are reinvested.
Value Growth Factor	For toggle notes, the factor by which the holding increases between the settlement date and the first cash pay date.

Term	Definition
Volatility	For floating rate notes, the relative change in the Discount Margin value for a 100 basis point change in the assumed rate. **Note:** The Duration Measurements for Italian Conventions appears for Italian floating rate notes. The gross and net duration and modified duration based on Italian market conventions.
vs	The corresponding benchmark. The drop−down menu displays a list of choices. You can save a custom benchmark for a bond.
Warrant	The right to purchase a fixed number of a company's stock at a specified price at any time during the exercise period. For this right, the buyer pays a premium over the current share price. The higher the premium, the more expensive the warrant. The cost of the right is substantially less than the cost of the share itself. Consequently, the price movements in the share are multiplied by the price movements of the warrant. This multiplier is the leverage or gearing factor. The higher the leverage, the cheaper the warrant. By comparing the premiums and gearings on a selection of warrants, you can assess their relative worth.
Warrant Parity	The price at which the percent premium is zero. Warrant parity is calculated as: ((stock price − exercise price) * # shares) / (par amount / 100)
Warrant Price	The price of the warrant as a percent of par.
Weighted Avg Cpn	The weighted average coupon for a security, calculated from weighting each bond's coupon by its par value as a percentage of the total par value of the fund.
Weighted Avg Life	For Danish mortgage bonds, the average number of years that each dollar of unpaid principal due remains outstanding.
Withholding Tax	For floating rate notes, the percentage of tax withheld.
Wkout	The workout date for the corresponding ask yield.
Wkout@ (price)	The price at the workout date.
Workout	The most likely redemption date for the security given the current price and redemption schedule.
Workout Date	For toggle notes, the default redemption date.
Workout Date Type	The most likely redemption date for the security, given the current price and redemption schedule.
Workout Yield	The yield at the workout date.
Worst Call	The date, price, and lowest calculated yield for all redemption dates.
XCCY - Cross Currency Benchmark Spread	A component available on the *Custom* tab that displays the spread of the fixed coupon to the selected same−currency benchmark bond. For example, when swapping a dollar issue into euro, it is the euro coupon equivalent of the dollar coupon spread to the default or selected euro benchmark bond.
YASA - YAS Activity	A component on the *Custom* tab that shows a log of your historical YAS calculations.

Term	Definition
Yield	The ask yield, which is the rate of return paid if the security is held to its workout date. The calculation is based on the coupon rate, length of time to maturity, and market price. The ask yield assumes that coupon interest paid over the life of the security is reinvested at the same rate. Yields are calculated to a specified workout type, date, and price. For more: Customizing Workout. • For South African governments, the annual interest rate divided by the market value. • For bonds in default, the purchase yield of the bond. • For toggle notes, the default yield of the bond. The assumptions underlying the default price/yield calculations are: the primary yield defaults to all cash yield when the last coupon payment was cash, or all pay-in-kind (PIK) yield when the last payment was made in kind. If the bond is in a PIK period, but is scheduled to pay cash starting from a specific date, the default price/yield calculations reflect the combination of PIK and cash pay. • On the *Yield to Call Analysis* (YTC) function, the yield to call.
Yield Curve	The type of price for the corresponding yield curve.
Yield Curve <FMC> #	For Brady bonds, the number and name of the yield curve used for the analysis.
Yield to Custom	The yield to a date and price you enter.
Yield To Maturity	The maturity date, price, and yield to maturity. Yield to maturity is the percentage rate of return paid if the security is held to its maturity date. The calculation is based on the coupon rate, length of time to maturity, and market price. Yield to maturity assumes that coupon interest paid over the life of the security is reinvested at the same rate.
Yield to Next Call	The next call date, price, and yield to the next call date.
Yield to Workout	For convertible bonds, appears for non-Japanese domestic convertibles only. The yield to maturity of the cash flows to the workout date.
Yield to Worst	The lowest yield that a buyer can expect among reasonable alternatives, such as yield to maturity, yield to call, and yield to refunding. If you change *Yield to Worst*, the yield is recalculated using the call date of the worst-to-yield at current market levels, but it may no longer be the worst.
Yield to Worst Call	The date, price, and yield of the lowest calculated yield for all redemption dates. If the yield to worst call is the same as the Strip Yield, "12Strip Yield" appears instead of the date and price.

Term	Definition
Yield Value of a (price change)	The change in yield of a security if price changes by a certain value. This is determined by calculating the difference between the yield to maturity and the yield if the bond price is increased by [x]. The lower the yield value of a [xx] price change the greater the dollar price volatility. The rationale is that it takes a smaller change in yield to produce a price change of [x]. For more: DUR Help Page. **Note:** Yield changes are almost symmetrical for small changes in price; therefore, it does not matter whether you increase or decrease price to calculate this value. For larger changes in price, there is a difference between the yield value if the price is increased or decreased. The change in yield value depends on the convexity of the security.
Yield W/ Inflation Assumption	• For Consumer Price Index (CPI) based floaters, the internal rate of return used to discount the projected nominal cash flows (the real plus inflation cash flows). • For non-U.S. inflation-indexed bonds, the nominal return. The return an investor would require on a normal bond to make the real return equal to that of the inflation-indexed bond, assuming a constant inflation rate.
Yield With Final Payment Bumped	Appears for Thai government bonds only. The yield where the maturity date is bumped if it falls on a weekend or holiday.
Yield Without Inflation	• For Consumer Price Index (CPI) based floaters, the internal rate of return used to discount the projected real cash flows. This yield is also interpreted as the return net of inflation. • For non-U.S. inflation-indexed bonds, the yield obtained after discounting the inflation assumption from the inflation-adjusted cash flows. This yield incorporates the three month lag impact and prior inflation adjustment made on the bond.
Yield-Beta Assumption	For non-U.S. inflation-indexed bonds, the slope of the relationship between nominal yield and real yields.
Yields are	The yield compounding frequency.
YIELD-with cash pay now	For pay-in-kind bonds, the yield, assuming the issuer pays cash for the current period, with the remaining periods paid as PIKs.
Yld Adv	The period of time you can recoup the premium paid for the bond, given the bond's yield advantage. For example: Yield Advantage = ([Points Premium/Bond Price]*100) / Current Yield Advantage.
Yld Drop (bp)	The yield drop, expressed in basis points and calculated as [(Interest Income – Finance Cost – Amortization) / (Risk Workout)]/100.

Term	Definition
Z-DM	The zero–discount margin, in basis points. The traditional analysis of floating rate notes revolves around the calculation of discount margin (DM). DM is an adjustment applied to the discount rate that equates the calculated price to the quoted price. On any reset date, if the DM equals the current spread or the spread scenario, the bond prices at par. If the issuer's credit has improved or deteriorated since the note was issued, the DM and the current spread or spread scenario diverge. To compute Z–DM, the implied zero curve is derived from the yield curve. Then the future values for the index are projected. These rates are used to imply the floating cash flows. Using these cash flows, the implied zero–rate for each payment date is obtained. These zero rates are used to discount the cash flows. Z–DM is the adjustment that must be applied to each zero rate so the modeled price matches the market.
Z-Sprd	The Z–Spread, which is the zero volatility spread of the selected bond. The Z–Spread is the selected bond's constant spread over the benchmark zero coupon swap curve.

Frequently Asked Questions

Get answers to the most commonly asked questions.

How do I calculate the yield to next call using a custom price?

You can calculate the yield to next call for a bond in the *Yield and Spread Analysis* (YAS) function and customize the variables used in the calculation, such as settlement date or price.

To calculate the yield to next call using a custom price:

1. Enter the bond ticker followed by YAS <GO>. For example, enter AN206325 <Corp> YAS <GO>.
2. In the *Price* field, enter a custom price.
3. From the drop-down menu next to the *Yield* field, select Next Call.
4. Press <GO>.

The *Yield* field updates to display the recalculated yield to next call.

For more on calculating yields: YAS Help Page.

You can also calculate the yield to next call using a custom price in Microsoft(R) Excel with a BDP override formula. You can use the YLD_YTC_ASK, YLD_YTC_BID, or YLD_YTC_MID field and the override YAS_BOND_PX.

For example:
=BDP("AN206325 Corp","YLD_YTC_ASK","YAS_BOND_PX=100")

For more: DAPI Help Page > Overrides Description.

What is the Bloomberg Generic (BGN) price for a bond?

Bloomberg Generic Price (BGN) is Bloomberg's market consensus price for corporate and government bonds. Bloomberg calculates the BGN price using a proprietary methodology, based on prices contributed to Bloomberg and other information Bloomberg considers relevant.

Bloomberg adjusts BGN methodology on an ongoing basis as needed to ensure it represents the market consensus. If a bond cannot be assigned an acceptable consensus price, it is marked "not priced." Bloomberg does not make a market in any securities covered by BGN.

To apply the BGN pricing source to an individual bond, enter the security ticker followed by PCS <GO> to launch the *Pricing Sources* function. For example, enter IBM 2.5 01/27/22 <Corp> PCS <GO>. You can also use PCS to apply BGN pricing to an entire class of bonds.

Once you apply the BGN price source to a bond, the BGN price appears in the quoteline and analytical functions across the Bloomberg Terminal®, such as DES <GO>, YAS <GO>, and GP <GO>.

For more on updating pricing source settings: PCS Help Page.

How is DV01 calculated?

DV01 is the dollar value change in market value given a one-basis-point change in yield (bonds), the underlying market swap curve used to value the swap (interest rate swaps and IR DV01 for CDS), or the par spread curve (Spread DV01 for CDS). Spread DV01 is calculated as the difference between the initial NPV of the CDS and the NPV after the par curve is shifted. Similarly, IR DV01 is calculated as the difference between the initial NPV of the CDS and the NPV after the swap curve is shifted.

- To calculate DV01 for bonds, enter YAS <GO> and use the *Yield & Spread Analysis* function. For more on DV01 for bonds, see the YAS Help Page.
- To calculate DV01 for interest rate swaps, enter SWPM <GO> and use the *Swap Manager* function. For more on DV01 for interest rate swaps, see the SWPM Help Page.
- To calculate Spread DV01 and IR DV01 for CDS, enter CDSW <GO> and use the *Credit Default Swap Valuation* function. For more on DV01 and IR DV01 for CDS, see the CDSW Help Page.

How can I convert a bond yield into another currency?

To convert the yield of a bond into another currency:

1. In the command line, enter the ticker followed by YAS XCCY <GO> to access the *Yield and Spread Analysis* function.
2. From the *Currency Pair* drop-down menu, select the currency.

What is the Z-DM for loans?

The traditional analysis of loans revolves around the calculation of discount margin (DM). DM is an adjustment applied to the discount rate that equates the calculated price to the quoted price. On any reset date, if the DM equals the quoted margin, the FRN prices at par. If the issuer's credit has improved or deteriorated since the loan was issued, the DM and the QM diverge.

To compute Z-DM, the implied zero curve is derived from the yield curve. Then the future values for the index are projected. These rates are used to imply the cash flows. Using these cash flows, the implied zero-rate for each payment date is obtained. The forward rates are used to discount the cash flows. Z-DM is the adjustment that must be applied to each forward rate so the modeled price matches the market. Z-DM does not account for premium calls.

What is the Z-Spread for a bond?

The Z-Spread, or zero volatility spread, is a bond's constant spread over the benchmark zero coupon swap curve, or the number of basis points to add to the spot curve in order to make the bond's net present value equal to its market price.

You can analyze a bond's Z-Spread in the *Yield and Spread Analysis* (YAS) and *Asset Swap Calculator* (ASW) functions. For more: YAS Help Page, ASW Help Page.

Take the next step.

For additional information, press the <HELP> key twice on the Bloomberg Terminal®.

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